The Importance of Screening and Early Identification of Depression in College Students

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Abstract

Background: The prevalence of depression among college students has significantly increased in the past decade. Depression is twice as common among college students than compared to the general population. Without early identification and treatment of depression, it can negatively impact important outcomes such as social connectedness, academic performance, academic retention, and future economic productivity.

Objective: Early identification and depression screening should be a priority among the college population. The purpose of this quality improvement project was to develop an evidence-based Depression Screening Protocol for an identified college health center utilizing the patient health questionnaire-2 (PHQ-2) and the patient health questionnaire-9 (PHQ-9) integrated in the electronic medical record. Various meetings with the healthcare providers, directors, and staff were held to share rationale for the development of the depression screening protocol and helped to identify the questions and concerns regarding implementation of the Depression Screening protocol.

Results: Of the college students receiving health services at the identified college health center, a total of 1,741 college students were screened for depression, 1,719 students completed the PHQ-2, 22 students completed the PHQ-9, and 13 students were referred to counseling and psychological services.

Conclusion: The Doctor of Nursing Practice (DNP) project demonstrated that the implementation of an evidence-based Depression Screening Protocol at a college health center was manageable and sustainable. This quality improvement project was successfully integrated into the identified health college center and it improved early identification of depression among college students.

Keywords: depression screening, PHQ-2, PHQ-9, college students, college health

The Importance of Screening and Early Identification of Depression in College Students

In the United States, more than 33.3% of first-year college students reported symptoms that are consistent with a diagnosable mental health disorder, according to the American Psychological Association (Auerbach et al., 2018). Mental health problems among college students have consistently increased over time and it poses a threat to their health and academic performance (Lipson, Lattie, & Eisenberg, 2019). Identification of mental health disorders among college students have been overlooked. Depression is the second most common mental health disorder among college students in the United States (Oswalt et al., 2018). College students are a key population for determining the economic growth and success of the nation. Thus, primary healthcare providers are the front line to early screening and identification of depression amongst college students.

Problem Statement

The prevalence of depression has risen in the college population in the past decade. There is an estimation of 33.3% of college students exhibiting significant symptoms of depression, generalized anxiety disorder, or suicide ideation (Oswalt et al., 2018). Depression is one of the most common mental health conditions among college students. In the United States, approximately 26% of individuals ages 18 and older live with a diagnosable mental health disorder in a given year (Oswalt et al., 2018). College students with depression can experience a decrease in physical health, negatively impact mental health, hinder relationships, impede academic endeavors, and decrease quality of life (Pedrelli, Nyer, Yeung, Zulauf, & Wilens, 2015).

Without proper management and treatment of depression, college students may participate in activities that could impact their well-being and could lead to substance abuse, self-

injury, and possibly suicide (Pedrelli el al., 2015). Mental illnesses such as depression has been associated with significant morbidity, mortality, disability, and increased risk of suicide (Khubchandani, Brey, Kotecki, Kleinfelder, & Anderson, 2016). Efforts to increase awareness and screening of depression among college students should be a priority.

Local Problem

An identified public college health center had a complaint-based approach to depression screening. A complaint-based approach screens for depression if a patient presented with depression-related symptoms. Patients may mask their depressive symptoms and suicidal intent by complaining of other health issues. Patients may report somatic symptoms in an attempt to avoid mental health stigma. Thus, the complaint-based approach may under-detect and under-treat depression.

The Patient Health Questionnaire-2 (PHQ-2) could be utilized as a first step approach to screen college students for depression during every visit at the college health center. The PHQ-2 has two questions and can easily be administered by healthcare providers at the college health center. College students who screen positive should be further evaluated with the PHQ-9 and possibly referred to counseling services at the identified public college. The counseling center at the identified public college provides an array of mental health services that could be beneficial for college students who are experiencing depression, but college students should first be screened and identified.

Clinical Question

In college-aged students (P), how does screening for depression with the Patient Health Questionnaire-2 (PHQ-2) (I) compared to current practice (C) affect early identification of depression (O)?

Specific Aims

The purpose of this Doctor of Nursing Practice (DNP) Project was to determine if the Patient Health Questionnaire-2 (PHQ-2) affects early identification of depression among college students.

Search Methods

A comprehensive literature search of electronic databases was conducted to identify existing, current literature regarding depression screening among college students. The following electronic databases were utilized: CINAHL, PubMed, ProQuest/Medline, ScienceDirect, Ovid, ProQuest Psychology Journals, and PsycARTICLES. Boolean search techniques were used to assist with the search methods, e.g., (depression AND college student) and (university student* OR college student* AND depressi* OR depressive symptom OR depressive disorder AND PHQ-2). Table A1 contains the literature search strategies (see Appendix A).

Inclusion criteria were: (1) participants consisted of college or university students of any age; (2) studies included depression and depressive symptoms; (3) studies containing the depression screening tool called the Patient Health Questionnaire (PHQ); (4) studies published as peer-reviewed; (5) the setting of the studies consisted of a college, university, or primary healthcare setting; (6) studies that included depression screening via the electronic medical record (EMR). Exclusion criteria were: (1) non-college students as participants; (2) languages other than English were restricted; (3) studies before the year 2014 were restricted; (4) studies without reports of depression or depressive symptoms (5) studies whose full texts could not be found.

Five (5) articles were requested from the FGCU Library ILLiad for full-text. The author of one (1) article was contacted via email to obtain the article in full-text. As listed in Table A2,

a total of 27 articles met the criteria for this review and each article was appraised using the Johns Hopkins Nursing Evidence-Based Practice Research and Non-Research Evidence Appraisal Tool. Table A2 outlines the Appraisal Table of the 27 articles (see Appendix A).

Available Knowledge/Review of the Literature

According to the U.S. Preventative Services Task Force (USPSTF), depression is among the leading causes of disability in individuals 15 years of age and older (Siu & U.S. Preventive Services Task Force (USPSTF), 2016). By 2020, depressive disorders are predicted to be the second leading cause of disability and death (Soltani, Smith, Beck, & Johnson, 2014). Prevalence of depression among college students accounts for approximately 16.7% and is more than twice as common among college students compared to the general population (Liu et al., 2019; English & Campbell, 2019). Thus, depression is highly prevalent and appears to be increasing. Depression is one of the leading health concerns in college population, yet roughly half of college students with depression receive treatment (English & Campbell, 2019). Mental health among college students affects important outcomes such as social connectedness, academic performance, academic retention, and future economic productivity, which should be a significant concern for certain stakeholders such as campus administrators, college professors, and parents (Lipson, Gaddis, Heinze, & Eisenberg, 2015; Ebert et al., 2018). In addition, depression has been associated with increased health care costs such as emergency department visits and inpatient hospitalizations (Wright et al., 2015). In the United States alone, the total cost of mental illness in young individuals is about $247 billion, which includes treatment costs, lost productivity, and crime (Cama et al., 2017).

The USPSTF recommends universal depression screening in all adults, regardless of risk factors, in all primary care settings (Siu & USPSTF, 2016). Unfortunately, the United States has

a complex mental healthcare system. There are several long-standing barriers to mental healthcare utilization: perceived stigma, personal stigma, lack of mental health literacy, fear of negative effect on academic record or job, limited access to mental health services, health insurance not covering treatment, unable to afford the out-of-pocket cost, fear of being committed or having to take medication, fear or mistrust of mental health professionals, and mental health professional shortage in the United States (Cadigan, Lee, & Larimer, 2019; Ennis et al., 2019; Kosyluk et al., 2016). In approximately 4,000 whole or partial U.S. counties, there are mental health professional shortages and some regions have less than one mental health professional per 30,000 people (Delaney, 2017). Mental healthcare facilities can be booked for weeks to months and have a waiting list, leading individuals to seek mental health services at acute settings and increasing emergency department visits, psychiatric hospitalizations, and risk for suicide (Lamsal, Stalker, Cait, Riemer, & Horton, 2018). Depression could lead to suicide, which is responsible for greater than 800,000 deaths annually worldwide (El-Den, Chen, Gan, Wong, & O'Reilly, 2018). On an economic standpoint, the mental healthcare demand exceeds the supply of mental healthcare services and providers.

In the literature review, there was a common theme of whether college health centers should implement universal depression screening or only screen high-risk students. Table 3A displayed common themes and categories of the literature review (see Appendix A). The debate arises from the number of students in need of mental health services greatly exceeds available mental health resources (Hill, Yaroslavsky, & Pettit, 2015). High-risk college students were categorized as students who were female, had baseline depression, neuroticism or psychoticism, social disconnection, negative feedback-seeking behaviors, negative automatic thoughts, negative rumination, dysfunctional attitude, sleep disturbances, stressful life events, prior

traumatic experiences (e.g. recent break-up with partner), prior suicidal behaviors, and childhood abuse (emotional, physical, neglect, or sex abuse) predict depressive symptoms (Liu et al., 2019; Hill, Yaroslavsky, & Pettit, 2015; Ebert et al., 2018; Çelik, Ceylan, Ünsal, & Çağan, 2019). Ideally, universal depression screening should be implemented at college entry to identify freshman or transfer students at high risk for depression, which may assist with the development and implementation of targeted preventive interventions (Ebert et al., 2018; Khubchandani et al., 2016). College students should be screened for depression at minimum once a year (Khubchandani et al., 2016). Overall, universal depression screening was preferred in contrast to only screening for at-risk college students (Khubchandani et al., 2016).

The most evidence-based option for college health centers would be to routinely screen for depression for each patient at every visit, regardless of referral question or presenting problem (English & Campbell, 2019; Maurer, Raymond, & Davis, 2018). There is a lack of evidence recommending a specific screening strategy, but a popular approach is to screen during routine health visits for the ease of implementation (Maurer, Raymond, & Davis, 2018; Manea et al., 2016). As stated previously, the USPSTF recommends routine depression screening provided that adequate systems are in place to ensure accurate diagnosis, effective treatment, and appropriate follow-up (Siu & USPSTF, 2016; Maurer, Raymond, & Davis, 2018; Wilson & Agius, 2017; English & Campbell, 2019). Routine systematic depression screening is supported by the USPSTF, American Academy of Pediatricians, the American Academy of Family Physicians, the American College of Preventive Medicine, and the World Health Organization (Siu & USPSTF, 2016; English & Campbell, 2019). Implementation of a routine depression screening protocol can be achieved through campus collaborations between academic affairs officials, student health centers, and counseling centers (Khubchandani et al., 2016). Colleges

should routinely screen for depression, provide technological low-cost interventions, and refer to college mental health services or external agencies when appropriate (Ennis et al., 2019). College health centers should routinely screen college students for depression to correspond with national evidence-based guidelines. Early identification and successful management of depression will lead to a significant improvement in depression severity (Soltani, Smith, Beck, & Johnson, 2015).

There are several valid and reliable depression screening tools available for healthcare providers. A systematic review of numerous depression screening tools in primary care suggested that the patient health questionnaire (PHQ) is the most extensively psychometrically tested tool (El-Den, Chen, Gan, Wong, & O'Reilly, 2018). The PHQ-2 is an abbreviated version of the PHQ-9 and comprised of two questions concerned about loss of interest or pleasure and feelings of depression and hopelessness over the past two weeks, consisting of the core symptoms of depression. The PHQ-2 appears to be the trend in depression screening because it takes less than two (2) minutes to administer and it can be quickly administered in a busy clinical setting (Wilson & Agius, 2017). The PHQ-2 makes it extremely appealing to busy clinical settings with time constraints, such as college health centers (Wilson & Agius, 2017). In order for successful implementation of an evidence-based practice change, screening instruments must be valid, reliable, brief, easy to administer, available at low-cost or free, and effective in achieving improved patient outcomes (Lakkis & Mahmassani, 2015). The PHQ-2 meets all of the criteria and should be utilized for initial screening in all age groups (Wright et al., 2016; Maurer, Raymond, & Davis, 2018; Wilson & Agius, 2017).

Another common theme of the literature review discussed the advantages and disadvantages of the different cut-off scores for the PHQ-2. The PHQ-2 is scored on a Likert

scale and the cumulative score is added together. There are two options regarding the cut-off score for the PHQ-2, depending upon the desired outcome. The meta-analysis by Manea et al. (2016) demonstrated that lowering the cut-off point of greater than or equal to (≥) two (2) will increase sensitivity at 0.91. Lowering the cut-off point ≥ 2 would lower the specificity to 0.7, given its inverse relationship with sensitivity (Manea et al., 2016). By increasing sensitivity, it may limit the number of people that would be missed by the depression screen. The cut-off score of ≥ 2 would be useful in screening at-risk populations because of the increased prevalence of depression (Manea et al., 2016). However, if the targeted population has a lower depression prevalence, it may result in an unacceptably high amount of false-positive screening rates. Based on the review of literature, the appropriate cut-off score was ≥ 3 to improve specificity over sensitivity (Slabaugh, Harris, & Wilcock, 2018; Manea et al., 2016; Wilson & Agius, 2017). Overall, the PHQ-2 is a simple, brief screening tool that can be utilized to detect depression in primary care settings, including college health centers.

The PHQ-2 is a standard of care that should be implemented as part of a systematic screen in order to reach its full potential and efficacy (Wilson & Agius, 2017). The literature review revealed that a systematic two-stage depression screening method yielded the best healthcare outcomes for patients. It was recommended to initially screen with the PHQ-2 and then further screen with the PHQ-9 if further evaluation was necessary (Smith, Brown Levey, & Lyon, 2017; Wright et al., 2016; Lakkis & Mahmassani, 2015; Soltani, Smith, Beck, & Johnson, 2015). Some studies had medical assistants, nursing staff, or clinic administrators screen college students with the PHQ-2 upon checking in for a visit or appointment (Smith, Brown Levey, & Lyon, 2017; Slabaugh, Harris, & Wilcock, 2018; Soltani, Smith, Beck, & Johnson, 2015).

Lastly, the most updated evidence suggested utilization of an electronic medical record (EMR) or electronic health record (EHR) for routine depression screening. In the era of technology, the healthcare system has adapted and merged with technology to become more efficient and advanced. Primary care settings without EMR systems have incorporated depression screening with the check-in process by printing out the PHQ-2 and attaching it to the registration paperwork. Other practices preferred to screen for depression via printed paper or EMR with the routine rooming process. The rooming process was comprised of medical personnel moving a patient from the waiting-room to an examination room. Once the patient was seated in the examination room, a medical staff member proceeded to record vital signs and update medical information in the patient's medical record.

Depression screening embedded into the EMR system could be utilized as a reminder to screen for depression at every visit. Thus, the PHQ scores could be saved automatically into the EMR system. As a result, having the depression screening protocol integrated into the EMR system could be time efficient and cost effective. On the contrary, the paper route could be time invasive due to printing out copies of the PHQ-2 and attaching it to the patient's physical chart. In addition, there could be a disadvantage due to the cost of office supplies such as paper and ink via the paper route. An advantage of depression screening interfaced with EMR was the feasibility of using the two-step screening process. The EMR system could prompt the PHQ-9 if the patient scores positively on the PHQ-2 and it could prompt follow-up recommendations for healthcare providers (Etter et al., 2017). Computer-based decision support systems focused on depression screening may inform the primary care practice of healthcare providers (Aalsma et al., 2018; Etter et al., 2017). One primary care clinic utilized an encounter decision aid (eDA) in the waiting room and the patients completed the PHQ-9 on an electronic eDA (Dannenberg et

al., 2019). According to the pilot study, the patients felt comfortable completing the PHQ-9 on the electronic eDA while in the waiting room (Dannenberg et al., 2019). An EMR system linked with depression screening may be more easily accessible and quicker for everyone.

Summary

The findings of the literature review recommended standardized depression screening among college students due to the high prevalence of depression and vulnerability of that population. The U.S. mental health system has long-standing barriers that affect access to mental health services and treatment. The USPSTF recommended standardized depression screening in all adults, regardless of risk factors. The literature demonstrated that the two-step depression screening process utilizing the PHQ-2 and PHQ-9 was the new gold standard and it could be implemented at any primary care setting, including college campuses.

Theoretical Framework

Rogers' Diffusion of Innovations Theory was chosen as a theoretical framework to guide the quality improvement project. In health care, change is inevitable and quality improvement projects fail due to various barriers such as poor planning, unmotivated staff, insufficient communication, or excessive changes (Barrow & Toney-Butler, 2019). The utilization of change theories could improve the odds of success and ensure effective change with practice improvement (Barrow & Toney-Butler, 2019). Change does not happen instantaneously but rather through a process in which will give people some time to adapt to the innovation. The main purpose of this project correlated with the last phase of this theory, which focused on adoption of the innovation and sustainability in practice.

Rogers' Diffusion of Innovations Theory explained change within a given system through five phases: (1) awareness, (2) interest, (3) evaluation, (4) trial, and (5) adoption (see

Appendix B). During the first phase, there was recognition or awareness of the need to change and knowledge of an innovation (Barrow & Toney-Butler, 2019; Orr & Davenport, 2015). Education and communication were needed during this phase to expose the staff for the need of change. The second phase involved persuading staff and piquing enough interest in the innovation, which allowed the change process to progress to the next phase (Barrow & Toney-Butler, 2019; Orr & Davenport, 2015). The third phase was an evaluation phase and the staff decided whether to accept or reject the change based on the merits of the innovation (Barrow & Toney-Butler, 2019; Orr & Davenport, 2015). If the staff decided to accept the innovation, the next phase of trial began by implementing new processes into practice and evaluating outcomes at intervals (Barrow & Toney-Butler, 2019; Orr & Davenport, 2015). The final phase of Rogers' Diffusion of Innovations Theory involved adoption or rejection of the innovation. Acceptance of the innovation would lead to continued use of the changed processes, which was the end goal for the DNP project.

Methods

Context

The identified college health center utilized a complaint-based screening approach as standard practice in which college students were only screened for depression if they presented with depression-related symptoms. The complaint-based screening approach was inconsistent with the United States Preventative Services Task Force (USPSTF) recommendation to screen every patient for depression at every visit, regardless of reason for the visit (Siu & USPSTF, 2016). Depression could prevent some students from seeking care, confirming the importance to follow the USPSTF recommendation for optimal detection of depression.

All college students with an appointment at the identified college health center were recruited for participation if they signed the consent form. Inclusion criteria included college students over the age of 18 years who presented to the identified college health center. The exclusion criteria included participants who were not enrolled at that identified college health center or were under the age of 18 years.

The setting of the quality improvement project was at an identified public college health center, which only serves the college student population. The healthcare providers within the college health center consisted of physicians and nurse practitioners. The other medical staff included registered nurses, licensed practical nurses, and medical assistants. The waiting room was separate from the exam rooms. The depression screening tool was administered in the exam rooms to facilitate confidentiality.

Setting

The project setting took place at an identified college health center at a public university consisting of approximately 15,000 students. All enrolled college students were eligible for healthcare services at the identified college health center, which was located on campus. The identified college health center utilized an EMR charting system called PyraMED. College students called or walked in for an appointment at the identified college health center. On-campus counseling services, professors, instructors, and/or staff could refer college students to the college health center as well. The medical staff at the college health center consisted of Medical Assistants (MAs), Licensed Practical Nurses (LPNs), Registered Nurses (RNs), Nurse Practitioners (NPs), and Medical Doctors (MDs).

Intervention

The healthcare providers and medical staff received an educational session of the Depression Screening Protocol prior to pilot period and before the two-month implementation period. The Depression Screening Protocol consisted of a two-step method involving the Patient Health Questionnaire-2 (PHQ-2) and the Patient Health Questionnaire-9 (PHQ-9). Figure C1 and Figure C2 contained the questions on both questionnaires (see Appendix C). A process map of the depression screening protocol was created for the ease of administration and implementation of the quality improvement project. Figure D1 displayed the Depression Screening Protocol for the pilot period (see Appendix D).

A pilot period was conducted at the identified college health center for one month. During the pilot period, eligible college students who met the inclusion criteria and who were present for a follow-up visit were given a consent form to participate in the DNP Project. If the student agreed, then the student was given the PHQ-2 by the following staff in the exam room: MA, LPN, or RN. The medical staff inserted the score into the electronic medical record (EMR) of the college student.

College students with a negative score on the PHQ-2 (less than 3) received a printed educational handout and the depression screening would be completed at that time. College students with a positive score on the PHQ-2 (equal to or greater than 3) were further screened by the healthcare provider. The healthcare provider would further screen the college student by administering question item nine (9) on the PHQ-9. Question nine (9) on the PHQ-9 determined if the student had suicide symptoms. If the response was yes, then the college student would immediately be referred to the counseling center at the identified public college. An immediate referral to the on-campus counseling center was considered a walk-in crisis. The on-campus counseling center collaborated with the college health center to create a walk-in crisis process if

a patient needed immediate mental health services. If the response to item nine (9) of the PHQ-9 was no, then the healthcare provider would have the option to complete the rest of the PHQ-9. If the healthcare provider decided to administer the remainder of the PHQ-9, a score of less than five (5) would warrant a printed educational handout. A score between five (5) and nine (9) and would warrant a printed educational handout, education by the healthcare provider during that visit, and would require a one (1) month follow-up. A score between five (5) and nine (9) would prompt the healthcare provider to give the student a printed educational handout, and the healthcare provider would have the option of non-urgently referring the student to the counseling center at the identified public college. A score greater than nine (9) would instruct the healthcare provider to give the student a printed educational handout and refer the student to the counseling center.

After the pilot period, data was collected for analysis. The data was de-identified and included the following information: the number of college students who were screened for depression, PHQ-2 scores, PHQ-9 scores, and the number of college students who were referred to the counseling center at the identified public college. In addition, the data included the age, gender, and ethnicity of the students who were screened with the Depression Screening Protocol. The principal investigator did not have access to the EMR system or other confidential information.

The purpose of the pilot period was to determine if the Depression Screening Protocol flowed smoothly and if adjustments needed to be made. The members of the college health center and on-campus counseling center had concerns about introducing an influx of referrals to the on-campus mental health services. The pilot period was conducted to ensure there were adequate mental health providers at the on-campus counseling center for the possibly influx of

patient referrals. The pilot period was trialed for one month from October 16, 2019 to November 13, 2019.

Once the pilot period was completed, the principal investigator had a meeting with the administrators of the college health center to discuss if the Depression Screening Protocol needed any adjustments. Since the pilot period was successful, the Depression Screening Protocol was adjusted to screen all students who presented to the college health center for the next two months. The depression screening protocol was administered to all eligible college students who met the inclusion criteria at every visit, except for laboratory visits, sexually transmitted infection testing visits, referred from the counseling center for medical evaluation, or if the PHQ screening was completed within the past two (2) weeks. Figure D2 illustrated the Depression Screening Protocol for the two-month implementation period (see Appendix D to compare the subtle changes made to the depression screening protocol for the two-month implementation period). After the completion of the two-month implementation period, the data was extracted in the same manner as the pilot period.

Data Collection

Data was collected from October 2019 to January 2020 via de-identified reports from the Director of the College Health Center. The de-identified reports contained the following information: demographic information such age, gender, and ethnicity; PHQ-2 scores; PHQ-9 scores; the number of college students who were screened; and the number of college students who were referred to counseling services. The PHQ-2 and PHQ-9 scores were entered into the EMR system of the college health center by the medical staff. The principal investigator did not have access to any student medical record. Involved medical staff received educational sessions regarding the Depression Screening Protocol, educational patient handout, and referral process

(see Appendix D for the Depression Screening Protocol and Appendix E for the educational handout).

Tools

This quality improvement project focused on two tools for depression screening: PHQ-2 and PHQ-9. The PHQ-2 was an abbreviated version of the PHQ-9 and comprised of two questions concerned about loss of interest or pleasure, and feelings of depression and hopelessness over the past two weeks. The PHQ-2 was utilized for initial screening and the PHQ-9 was used for further evaluation for a positive PHQ-2 score. The administration of the PHQ-2 took less than two (2) minutes to complete, which was beneficial at a busy college health center (Wilson & Agius, 2017). The PHQ-2 scores ranged from zero (0) to three (3), with zero (0) being not being depressed at all and three (3) being feeling depressed nearly every day (see Appendix C).

The PHQ-2 had a sensitivity nearing 100% and specificity of 77% for detecting depression (Maurer, Raymond, & Davis, 2018). The PHQ-2 was utilized as a pre-screener to the PHQ-9, which consisted of nine (9) items (see Appendix C). A total PHQ-9 score of 10 or higher had a sensitivity of 89.5% and a specificity of 72.1% for detecting depression (Wright et al., 2016). The PHQ-2 had a sensitivity comparable with the PHQ-9 (Maurer, Raymond, & Davis, 2018). The PHQ-2 and PHQ-9 were valid, reliable, brief, easy to administer, available at low-cost or free, and effective depression screening tools (Lakkis & Mahmassani, 2015). Both screening tools may be utilized for any age group (Wright et al., 2016; Maurer, Raymond, & Davis, 2018; Wilson & Agius, 2017). It is important to acknowledge that the PHQ-2 and PHQ-9 were useful tools for screening purposes and not for diagnostic purposes.

The Depression Screening Protocol was created based on the USPSTF and the American College Health Association (ACHA) guidelines with an interdisciplinary collaboration of the Medical Director of the college health center, Director of the college health center, DNP Project Chair, and a Doctoral Psychology Intern at the on-campus counseling center. The purpose of the protocol was to develop a routine systematic screening for depression and to identify students with depressive and/or suicide symptoms. A process map of the Depression Screening Protocol was developed to facilitate the healthcare providers with a systematic approach to early identification of depression among college students, to provide them with educational material and to refer for psychological services if needed. Educational handouts were created for college students whom scored five or higher on the PHQ-9 or whom responded "Yes" to item number nine (9) on the PHQ-9. The educational handout contained information about depression symptoms, interventions, and where to go for assistance. Figure E1 and E2 contains the front and back page of the educational handout (see Appendix E). This protocol utilized evidence-based practice guidelines to screen, identify, and refer college students with an interdisciplinary collaboration between the college health center and the on-campus counseling center.

Outcome Measures and Analysis

The outcome measures of this quality improvement project included the following: the percentage of college students initially screened with the Patient Health Questionnaire-2 (PHQ-2), the percentage of college students who scored positively on the PHQ-2, the percentage of college students screened with the PHQ-9, the percentage of college students who scored positively on the PHQ-9, and the percentage of college students referred to counseling and psychological services. The data was analyzed with the Statistical Package for Social Sciences (SPSS) IBM Version 26.0 for Microsoft Windows.

Ethical Considerations

The quality improvement project was a systematic, evidence-based practice protocol for earlier identification of depression among college students. This project pursued to provide a standard of care at the identified health college center and followed established ethical guidelines pertaining to the protection of the college students. The college students' medical records were kept confidential and the principal investigator did not have access to the electronic medical record (EMR).

The principal investigator only had access to de-identified reports, which could not be traced back to a specific student. This project met the requirements of confidentiality and Health Insurance Portability and Accountability Act (HIPPA). The principal investigator consulted the Institutional Review Board (IRB) and received ethical approval to implement this DNP project at the identified health college center. The principal investigator confirmed that the project presented in this manuscript met the ethical guidelines, including adherence to the legal requirements, of the United States and received approval from the IRB from the participating institution.

Results

The purpose of this quality improvement project was to assess current depression screening practices at an identified college health center and develop an evidence-based depression screening protocol to identify depression among college students. A needs assessment discovered that the identified college health center utilized a complaint-based approach to screening for depression. Current evidence recommended screening each patient for depression at every visit, regardless of referral question or presenting problem. The evidence-based literature review revealed that screening college students in a systematic, two-stage method

yielded the best patient outcomes. The two-stage depression screening protocol was comprised of initially screening with the PHQ-2 and then further screening with the PHQ-9 if further evaluation was necessary. PHQ-9 was primarily utilized to rule out suicidal ideation. Positive depressive symptoms warranted a patient handout that listed mental health resources, mental health referral, and follow-up visit based on the outcomes documented by above mentioned tools. The pilot period was conducted from October 16, 2019 through November 13, 2019. The two-month implementation period was completed from December 2, 2019 through January 31, 2020. The final results and findings of the DNP Project were presented to the college health center on March 13, 2020.

Educational sessions via a brief PowerPoint presentation were developed and approved by the IRB. A consent form, an evidence-based depression screening protocol, and a patient handout were approved by the IRB. Educational sessions were scheduled prior to the pilot period and prior to the two-month implementation period. The educational sessions were presented to the college health center on September 23, 2019 and November 23, 2019. The educational sessions were presented to the healthcare providers and medical staff of the identified college health center and all questions were answered appropriately. The educational sessions increased awareness of the evidence-based depression screening protocol, screening tools, printed patient handout, and the referral process. Specific instructions and screening forms were discussed and reviewed regarding the role of the healthcare providers and the medical staff within the Depression Screening Protocol. The consistency, standardization, and sustainability of the Depression Screening Protocol was emphasized throughout the educational presentation. The educational PowerPoint presentation may be edited to update statistics and update documents within the Depression Screening Protocol if needed in the future. The ultimate goal of the

educational sessions focused on increased awareness of the Depression Screening Protocol by the college health center for earlier detection and treatment of depression among college students.

The Depression Screening Protocol was integrated into the EMR system of the college health center for ease of administration. The healthcare providers and medical staff were not required to memorize the screening questions because the EMR system was modified to include the following: PHQ-2 questionnaire, PHQ-9 questionnaire, PHQ scores, and EMR referral codes (CAPS Nonurgent and CAPS Crisis). The Depression Screening Protocol was embedded into the EMR system and reminded medical staff and healthcare providers to screen for depression at every visit. Additionally, PHQ scores were automatically saved into the patient's medical record and embedded into the progress note for that encounter date. This specific process saved time, decreased medical errors, and conserved office supplies such as paper, ink, and staples.

Results were aggregated utilizing Microsoft Excel and then imported into the SPSS IBM Version 26.0 for Microsoft Windows. The data analysis focused on descriptive statistics including frequencies of PHQ-2 scores, PHQ-9 scores and referrals to counseling and psychological services. That analysis was then compared across various demographic measures such as age, gender, and ethnicity. PHQ-2 scores ranged from zero (0) to six (6) and PHQ-9 scores ranged from zero (0) to 21.

The DNP project significantly increased the quantity of students being screened for depression. 210 students were screened with the PHQ-2 and 5 students were further screened with the PHQ-9 during the pilot period (Table 1). Thus, the total sample size of the pilot period was comprised of 215 students. The pilot period yielded 11 positive PHQ-2 scores (≥ 3) and were further screened with the PHQ-9. Six (6) participants were provided a patient handout with

mental health resources and were referred non-urgently to counseling services (Table 2). The pilot period did not yield any students who required a crisis walk-in referral to the on-campus counseling center. During the pilot period, 75.8% of students scored zero (0), 6% of students scored one (1), and 10.7% of students scored two (2), indicating that 92.5% of students (n = 215) scored negatively (Table 3).

Table 1 *Pilot Period PHQ-2 & PHQ-9 Frequencies*

	Frequency	**Percent**	**Valid Percent**	**Cumulative Percent**
PHQ-2	210	97.7	97.7	97.7
PHQ-9	5	2.3	2.3	100.0
Total	215	100.0	100.0	

Table 2 *Pilot Period Referrals*

	Frequency	**Percent**	**Valid Percent**	**Cumulative Percent**
CAPS Nonurgent	6	100.0	100.0	100.0

Table 3 *Pilot Period PHQ Scores*

	Frequency	**Percent**	**Valid Percent**	**Cumulative Percent**
0	163	75.8	75.8	75.8
1	13	6.0	6.0	81.9
2	23	10.7	10.7	92.6
3	4	1.9	1.9	94.4
4	5	2.3	2.3	96.7
6	3	1.4	1.4	98.1
8	1	.5	.5	98.6
12	1	.5	.5	99.1
15	1	.5	.5	99.5
21	1	.5	.5	100.0
Total	215	100.0	100.0	

During the pilot period, 94.8% of the students (n = 215) screened with the Depression Screening Protocol were between the ages of 18 through 22 (Table 4). Roughly two-thirds (67.9%) of the pilot period sample (n = 215) were female and less than one-third (31.2%) were male (Table 5). Less than 1% of the screened students chose not to report their gender (see Appendix F). Regarding the ethnicity of the participants, approximately 40.5% were White Non-Hispanic, 14% were Hispanic, 7% were African American, and 2% were Asian (Table 6). Unfortunately, almost one-half (48.9%) of students (n = 215) did not report their ethnicity (see Appendix F).

Table 4 *Pilot Period Age Results*

	Age	**Frequency**	**Percent**	**Valid Percent**	**Cumulative Percent**
	18	48	22.3	22.4	22.4
	19	54	25.1	25.2	47.7
	20	45	20.9	21.0	68.7
	21	31	14.4	14.5	83.2
	22	26	12.1	12.1	95.3
	23	3	1.4	1.4	96.7
	26	1	.5	.5	97.2
	27	2	.9	.9	98.1
	28	2	.9	.9	99.1
	29	1	.5	.5	99.5
	35	1	.5	.5	100.0
	Total	214	99.5	100.0	
Not Reported	**-**	1	.5		
Total		215	100.0		

Table 5 *Pilot Period Gender Results*

Gender	Frequency	Percent	Valid Percent	Cumulative Percent
Female	146	67.9	67.9	67.9
Male	67	31.2	31.2	99.1
Not Reported	2	.9	.9	100.0
Total	215	100.0	100.0	

Table 6 *Pilot Period Ethnicity Results*

	Frequency	Percent	Valid Percent	Cumulative Percent
African American-Non-Hispanic	7	3.3	3.3	3.3
Asian/Pacific Islander	2	.9	.9	4.2
Hispanic	14	6.5	6.5	10.7
No Indication / Not Reported	93	43.3	43.3	54.0
Not Reported	12	5.6	5.6	59.5
White Non-Hispanic	87	40.5	40.5	100.0
Total	215	100.0	100.0	

After the completion of the pilot period, the identified college health center transitioned into the two-month implementation period. The total sample of the two-month implementation period was comprised of 1,526 students (see Appendix G). 1,509 students were screened with the PHQ-2 and 17 students were further screened with the PHQ-9 (Table 7). The two-month

implementation period resulted in 63 positive PHQ-2 scores (≥ 3) and were further screened with the PHQ-9. During the two-month implementation period, 81.4% of students scored zero (0), 8.2% of students scored one (1), and 5.8% of students scored two (2), indicating that 95.4% of students (*n* = 1526) scored negatively (Table 8). Seven (7) participants were provided an educational handout and were non-urgently referred to counseling services (Table 9). The two-month implementation period did not require a crisis walk-in referral for any participants.

Table 7 *Two-Month Implementation Period PHQ-2 & PHQ-9 Frequencies*

	Frequency	**Percent**	**Valid Percent**	**Cumulative Percent**
PHQ-2	1509	98.9	98.9	98.9
PHQ-9	17	1.1	1.1	100.0
Total	1526	100.0	100.0	

Table 8 *Two-Month Implementation Period PHQ Scores*

	Frequency	**Percent**	**Valid Percent**	**Cumulative Percent**
0	1242	81.4	81.4	81.4
1	125	8.2	8.2	89.6
2	89	5.8	5.8	95.4
3	25	1.6	1.6	97.1
4	27	1.8	1.8	98.8
5	6	.4	.4	99.2
6	6	.4	.4	99.6
7	2	.1	.1	99.7
8	1	.1	.1	99.8

11	1	.1	.1	99.9
13	1	.1	.1	99.9
14	1	.1	.1	100.0
Total	1526	100.0	100.0	

Table 9 *Two-Month Implementation Period Referrals*

	Frequency	**Percent**	**Valid Percent**	**Cumulative Percent**
CAPS Nonurgent	7	100.0	100.0	100.0

During the two-month implementation period, 89% of the students (*n* = 215) screened with the Depression Screening Protocol were between the ages of 18 through 22 (Table 10). Roughly three-quarters (71.7%) of the two-month implementation period sample (*n* = 215) were female and about one-fourth (28.3%) were male (Table 11). Regarding the ethnicity of the participants, approximately 34.1% were White Non-Hispanic, 8% were Hispanic, 4% were African American, 1.4% were Asian, and less than 0.1% were American Indian or Alaskan Native (Table 12). Unfortunately, about one-half (52.3%) of students (*n* = 215) did not report their ethnicity (see Appendix G). Figure G6 illustrated the frequency of encounter dates for depression screening. Thus, the healthcare providers and medical staff were consistent with screening at every visit during the two-month implementation period (see Appendix G). The end of December 2019 through January 6, 2020 depicted less encounter dates for depression screening because the last day of the fall semester was December 14, 2019. The Spring 2020 semester started January 6, 2020 and Figure G6 illustrated that increase in depression screening (see Appendix G).

Table 10 *Two-Month Implementation Period Age Results*

Age	Frequency	Percent	Valid Percent	Cumulative Percent
18	198	13.0	13.0	13.0
19	357	23.4	23.4	36.4
20	327	21.4	21.4	57.8
21	304	19.9	19.9	77.7
22	172	11.3	11.3	89.0
23	57	3.7	3.7	92.7
24	29	1.9	1.9	94.6
25	20	1.3	1.3	95.9
26	7	.5	.5	96.4
27	4	.3	.3	96.7
28	4	.3	.3	96.9
29	10	.7	.7	97.6
30	3	.2	.2	97.8
31	4	.3	.3	98.0
32	5	.3	.3	98.4
35	1	.1	.1	98.4
36	1	.1	.1	98.5
37	5	.3	.3	98.8
38	1	.1	.1	98.9
39	4	.3	.3	99.1
43	1	.1	.1	99.2
45	1	.1	.1	99.3
47	1	.1	.1	99.3
48	2	.1	.1	99.5
51	2	.1	.1	99.6
54	5	.3	.3	99.9
59	1	.1	.1	100.0
Total	1526	100.0	100.0	

Table 11 *Two-Month Implementation Period Gender Results*

Gender	Frequency	Percent	Valid Percent	Cumulative Percent
Female	1094	71.7	71.7	71.7
Male	432	28.3	28.3	100.0
Total	1526	100.0	100.0	

Table 12 *Two-Month Implementation Period Ethnicity Results*

	Frequency	Percent	Valid Percent	Cumulative Percent
African American-Non-Hispanic	61	4.0	4.0	4.0
American Indian/Alaskan Native	2	.1	.1	4.1
Asian/Pacific Islander	21	1.4	1.4	5.5
Hispanic	122	8.0	8.0	13.5
No Indication / Not Reported	695	45.5	45.5	59.0
Not Reported	104	6.8	6.8	65.9
White Non-Hispanic	521	34.1	34.1	100.0
Total	1526	100.0	100.0	

Discussion

Overall, a total of 1,719 college students were screened with the PHQ-2 and 4.3% scored positively on the PHQ-2 (≥ 3). Thus, 95.7% (*n* = 1,719) of college students scored negatively on the PHQ-2. A total of 22 students were screened with the PHQ-9 and 27.2% scored positively on the PHQ-9 (> 9). Therefore, 72.8% of college students scored negatively on the PHQ-9. The healthcare providers stated that most students with positive PHQ-9 scores were already undergoing depression counseling and treatment. In summary, 1,741 students in total were

screened with the Depression Screening Protocol and only 0.7% needed a referral to counseling and psychological services.

In this small sample (n = 1,741) of college students, 27.2% scored positively on the PHQ-9, which compares to a larger study (n = 155,026) reporting 26.9% screening positive for depression (Lipson, Lattie, & Eisenberg, 2019). The national rate of depression in college students was estimated around 30-36% (Slabaugh, Harris, & Wilcock, 2018). The findings of this DNP project revealed that the rate of depressive symptoms in the identified college health center was around 27.2%, which is less than the national rate. Despite the low detection rate of depressive symptoms, the continuation of this Depression Screening Protocol is necessary to prevent the potential negative consequences of undetected and untreated depression. Even the prevention of one suicide makes the effort meaningful.

As stated previously, the Rogers' Diffusion of Innovations Theory guided this DNP project. The five phases of the theory included awareness, interest, evaluation, trial, and adoption. During the first phase, the principal investigator provided educational sessions to the college health center to increase awareness of the need to change. Education and communication were key components to increasing awareness of the need to change among the healthcare providers and medical staff at the identified college health center. The second phased involved persuading the healthcare providers and medical staff of the need to change their approach to depressions screening. The third phase consisted of evaluating the feedback from the college health center on whether they would accept or reject the need to change. Once they accepted the need to change, the principal investigator suggested a pilot period to assess the new protocol and evaluate the outcomes at intervals. The members of the college health center agreed that the trial phase would be beneficial and everyone accepted to try the new Depression Screening Protocol.

The last phase of the Rogers' Diffusion of Innovations Theory involved adoption or rejection of the innovation. Fortunately, the college health center accepted the Depression Screening Protocol and continued to utilize the changed processes as their standard of care, which was the ultimate goal of the DNP project.

Following the completion of the DNP project, the results were presented to the healthcare providers and medical staff at the college health center. The healthcare providers and medical staff found the Depression Screening Protocol to be brief, user-friendly, reliable, valid, cost effective, and improved patient outcomes. The identified college health center commented that the PHQ screening integrated into the EMR system was helpful and time effective. The staff from the identified college health center discussed that the college students responded positively to the Depression Screening Protocol. Some students voiced to the healthcare providers that they were grateful for the screening because they had a loved one or friend with depression which resulted in suicide. Overall, the Depression Screening Protocol received positive feedback from the healthcare providers, medical staff, and college students, and it proved to be sustainable with no major difficulties with the implementation.

Project Strengths

This DNP Project resulted in the development of an Evidence-Based Depression Screening Protocol focused on standardized depression screening of college students over the age of 18 years at an identified college health center. The first strength of this project coincided with developing an evidence-based protocol for this college health center. Secondly, embedding the protocol into the EMR system was key to making this quality improvement project sustainable. Despite the completion of the DNP project, the college health center continues to utilize the Depression Screening Protocol as a standard of care.

Subsequently, the educational sessions increased the awareness of the healthcare providers and medical staff regarding the importance of screening depression as a standard of care. The educational sessions had positive outcomes because it allowed the principal investigator to build rapport with the college health center and receive constructive feedback. Additionally, this DNP Project contributed to the collaboration between the college health center and the on-campus counseling center, which improved the referral process and enhanced continuity of quality care. Improved communication between the college health center and the on-campus counseling center assisted the college students in receiving appropriate mental health resources in a timely fashion. Thus, closing the gap and not missing students in need of mental health services.

Lastly, various students reported positive comments to the college health center regarding the new Depression Screening Protocol and they were thankful to be heard and acknowledged. Prior to implementing this DNP Project, some healthcare providers voiced their concerns about the students responding negatively to the Depression Screening Protocol and possibly causing an influx of referrals to the on-campus counseling center. To their surprise, the students had the opposite reaction to the healthcare providers' assumptions and the total results revealed zero (0) students referred as a crisis walk-in to the on-campus counseling center. Overall, the DNP project was successful, feasible, and sustainable.

Interpretation

The results of this DNP Project corresponded with the positive changes as seen in the literature review (Smith, Brown Levey, & Lyon, 2017; Wright et al., 2016; Lakkis & Mahmassani, 2015; Soltani, Smith, Beck, & Johnson, 2015). The literature review found that a two-stage depression screening protocol improved the screening process at college health

centers. Additionally, the literature recommended screening for depression for each patient at every visit, regardless of referral question or presenting problem (English & Campbell, 2019; Maurer, Raymond, & Davis, 2018).

This DNP Project changed the college health center's method of depression screening from a complaint-based screening approach to a universal standard of care screening approach. This evidence-based approach to screening increases the chance of earlier identification of depression among college students. Thus, providing needed mental health resources to college students presenting with depressive symptoms and/or suicide ideation. These improved practices may improve academic retention, improve academic performance, and decrease suicide rates. Overall, the Depression Screening Protocol improved the screening process at the identified college health center, provided mental health resources, and referred students to counseling services when appropriate, which is in compliance with national evidence-based guidelines.

Implications

Practice

The impact of depression screening goes beyond the college health center. The collaboration between the college health center and counseling centers affect the quality and frequency of mental health services provided to the college students. Fluid communication between those centers could close the gap between student needs and available resources. On-campus counseling centers may not have appointments available for up to one month, which could delay the treatment of depressive symptoms. Considerations for integration of counseling services within the college health center and outsourcing to off-campus counseling centers could assist with the growing mental health needs of college students.

The DNP project has resulted in the development of an evidence-based Depression Screening Protocol for college health centers. A relationship with a local, off-campus counseling center was established during the implementation of the DNP project. The contact information of the off-campus counseling center was given to the director of the identified college health center in hopes of partnering in the near future. The off-campus counseling center offered availability to accommodate the students' schedule and schedule appointments within one to three days. Lastly, the off-campus counseling center offered telehealth services, which would connect a mental health professional to a college student via a live video connection utilizing a smartphone or tablet. Mental health services delivered via telehealth could be beneficial to college students in need of immediate mental health services, especially during the current pandemic of the 2019 novel coronavirus (COVID-19).

Research

Based on the number of positive PHQ-2 and PHQ-9 scores, future studies should be conducted to determine the compliance rates of follow-up visits. Students with high scores are strongly encouraged to schedule a follow-up appointment to determine if they have received mental health services in a timely manner and if the plan of treatment was effective. Adherence to follow-up visits were not tracked and future investigation should focus on the percentage of students who attended the follow-up appointment.

Some healthcare providers from the identified college health center discussed that students were already receiving interventions for depression. This project did not collect data regarding the participants' medical history, medication list, or if they were already receiving mental health interventions. Future studies should concentrate on the participants' medical

history, current medication list, and psychological history to determine high-risk factors associated with depressive symptoms.

Future investigation should concentrate on closing the gap via fluid communication and partnerships between college health centers and counseling centers. Development of evidence-based mental health programs and online mental health services in conjunction with professional support through counseling centers should be explored as ways to engage college students while minimizing burden on the healthcare delivery systems.

Potential Influence on Current Practice / Health Policy

Future projects should address the follow-up gap to ensure that college students receive mental health services in a timely manner. Future projects should address the health policy needs, which include an assessment of local mental health programs and strategies at the collegiate level to create collaborations in order to provide evidence-based mental health interventions. Effective relationships with on-campus counseling centers are vital to establish fluid communication regarding the optimal plan of treatment for students with depressive symptoms. Establishing partnerships with off-campus counseling centers could result in time-effective appointments, increased compliance, and improved follow-up visits. Additionally, this project may assist in advocating for the increased numbers of counselors on college campuses and college health centers, which would increase access to care for students.

Limitations

Although this quality improvement project revealed positive health outcomes, the limitations of this project should be considered when drawing conclusions. First, the PHQ-2 and PHQ-9 questionnaires utilized in the Depression Screening Protocol required the participants to self-report the scores. Self-reporting could contribute to self-reporting bias and factual error.

Almost half of the students screened with the Depression Screening Protocol did not report their ethnicity, which could skew the results and may not represent the general population.

Subsequently, this project only included college students over the age of 18 years at a small college health center. There were students under the age of 18 who visited the college health center that were not accounted for due to the inclusion and exclusion criteria. A small sample size and one setting may not represent the average college population.

Lastly, this project did not collect data regarding pre-existing medical and psychological diseases, list of medications, or if the participants were already undergoing counseling and psychological services. That information would be useful to know if the students' mental health diagnoses were under control or if they needed more mental health resources. A similar project could assess the generalizability and feasibility of this Depression Screening Protocol at other larger-scaled colleges for earlier identification of depression among college students.

Conclusions

Overall, the results and findings of the DNP project revealed that the implementation of an Evidence-Based Depression Screening Protocol at a college health center was successful, sustainable, and tailored to identify depression among college students. The participants reported positive comments to the healthcare providers and medical staff regarding the new Depression Screening Protocol. The healthcare providers and medical staff reported increased awareness of the importance of screening for depression among college students at every visit. Generally, the impression of the Depression Screening Protocol was highly positive and the proposed changes were readily accepted by all staff from the identified college health center. Early identification remains an important component of screening, management, and prevention of further complications of untreated depression such as suicide. The Depression Screening Protocol may

demonstrate increased safety and quality of care of college students by identifying depressive symptoms and referring to counseling earlier. The DNP project contributed to the literature by demonstrating that the Depression Screening Protocol had a positive impact by creating a standardized, feasible screening method to identify depression among college students.

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Appendix A

Table A1: Search Strategies

Literature Search of Screening and Early Identification of Depression in College Students

Date	Database	Keywords/Synonyms Phrase	Search Strategy*	Yield **
2/2/19	CINAHL	depression AND college students	Peer reviewed and published from 2014-2019	817/5
2/2/19	CINAHL	depression AND college students	Peer reviewed and published from 2014-2019 in Full-Text	296/1
2/23/19	CINAHL	depression AND college students	Peer Reviewed, 2014-2019, USA only, English Language Only	473/1
2/23/19	Back-sourced	None	N/A	1/1
3/23/19	CINAHL	depression AND college students	Peer Reviewed, 2014-2019, USA only, English Language Only	8/1
3/23/19	CINAHL	PHQ-2 AND college students	Peer Reviewed, 2014-2019, USA only, English Language Only	0/0

Notes. CINAHL (Cumulative Index of Nursing and Allied Health Literature);
*Include peer-reviewed journals, hand searing, Boolean operators, wildcards, and truncation
**Original yield / full text yield

3/24/19	CINAHL	PHQ-2 AND depression screening	Peer Reviewed, 2014-2019, USA only, English Language Only	20/3
4/6/19	PubMed	university student* OR college student* AND depressi* OR depressive symptom OR depressive disorder AND PHQ-2	Peer reviewed and published from 2014-2019 in English only	95,004/0
4/6/19	PubMed	university student* OR college student* AND depressi* OR depressive symptom OR depressive disorder AND PHQ-2	Peer reviewed, 2014-2019, English only, default sort order to 'Best Match' instead of 'Most Recent'	176,625/1
4/6/19	PubMed	university student* OR college student* AND PHQ-2	Peer reviewed, 2014-2019, English only, default sort order to 'Best Match' instead of 'Most Recent'	8/2
4/12/19	ProQuest (Medline)	PHQ-2 AND depression screening	Peer reviewed and published from 2014-2019 in English only	3/0
4/12/19	ProQuest (Medline)	PHQ-2 AND depression	Peer reviewed and published from 2014-2019 in English only	6/0

4/12/19	ProQuest (Medline)	depression screening AND college students	Peer reviewed and published from 2014-2019 in English only	3/1
4/15/19	ScienceDirect	PHQ-2 AND college students OR university students	Published from 2014-2019	606/1
4/15/19	ScienceDirect	PHQ-2 AND college students AND depression	Peer reviewed and published from 2014-2019	65/1
4/15/19	ScienceDirect	PHQ-2 AND college OR university AND depression screening	Published from 2014-2019	989/0
4/19/19	Ovid	depression screening AND college students	Peer reviewed and published from 2014-2019	5/2
4/23/19	ProQuest Psychology Journals	university student* OR college student* AND PHQ-2 AND depression screening	Peer reviewed and published from 2014-2019 in English only	78,029/0

4/23/19	ProQuest Psychology Journals	university student* OR college student* AND PHQ-2 AND depression screening	Peer reviewed and published from 2014-2019 in English only; United States only	7,515/0
4/23/19	ProQuest Psychology Journals	depression screening AND college student	Peer reviewed and published from 2014-2019	6,412/1
4/23/19	PsycARTICLES (Proquest)	depression screening AND college student	Peer reviewed and published from 2014-2019 in English only	6,387/2
4/23/19	PsycARTICLES (Proquest)	depression screening AND college student AND PHQ-2	Peer reviewed and published from 2014-2019 in English only	83/0
5/6/19	ScienceDirect	PHQ-2 AND EMR	Peer reviewed and published from 2014-2019	96/1
5/6/19	ScienceDirect	PHQ-2 AND EMR AND college OR university	Peer reviewed and published from 2014-2019	47/1
5/6/19	ScienceDirect	electronic health record OR EMR OR electronic medical record AND PHQ-2	Peer reviewed and published from 2014-2019	93/1

5/6/19	ScienceDirect	nurse practitioner* AND depression screening and PHQ-2	Peer reviewed and published from 2014-2019	263/1
5/6/19	ScienceDirect	Improving depression screening among college students	Peer reviewed and published from 2014-2019	2,982/1
5/10/19	CINAHL	universal depression screening	Peer reviewed and published from 2014-2019	55/1
5/10/19	PubMed	PHQ-2 and screen* for depression	Peer reviewed and published from 2014-2019	124/1

Table A2: Appraisal Table

John Hopkins Nursing Evidence-Based Practice

Article Number	Author and Date	Evidence Type	Sample, Sample Size, Setting	Findings That Help Answer the EBP Question	Observable Measures	Limitations	Evidence Level, Quality
1	(Yan, L., Ning, Z., Guangyi, B., Yubei, H., Bingyuan, J., Yili, W., Chuanxin, L., & Gongying, L., 2018)	Systematic Review & Meta-Analysis of Cohort Studies	24 cohort studies N = 25,154 college students in China, United States, Japan, Spain, Taiwan, Switzerland, and Germany	This systematic review discusses predictive factors for depressive symptoms among college students. The following factors contribute to depression: biological, psychological, and environmental factors. This study improves understanding of depressive symptoms.	Outcome measures were evaluated using the Stata version 13.1 and Microsoft Excel 2013. Sensitivity analyses were conducted, and a funnel plot of each factor was made.	Limited databases were searched for literature collection. There were no languages and publication period restrictions. Different screening methods were utilized.	Level IIA High Quality

2	(Çelik, Ceylan, Ünsal, & Çağan, 2019)	Single Descriptive Study	N = 445 college students who were studying at Faculty of Health Sciences at Eskisehir Osmangazi University in Turkey	Approximately one-fifth of college students demonstrated depressive symptoms and mild depressive symptoms. There was no statistical difference in the frequency of depressive symptoms between gender. College students who experienced academic failure, low level of income, smoke use, and alcohol use showed more higher depressive symptoms. Students who had poor quality sleep had a higher risk of having depressive symptoms.	The Beck Depression Inventory and the Pittsburg Sleep Quality Index questionnaire were utilized as data collection tools. Pre-prepared questionnaire forms were filled by students under observation. SPSS was used for the data analysis.	Sample was not randomly selected. Convenience sample. Sample not representative of the general population in the United States.	Level VI Good Quality

3	(Khubchandani, Brey, Kotecki, Kleinfelder & Anderson, 2016)	Single Descriptive Study	N = 934 Undergraduate students at a Midwestern university	PHQ-2 had a high reliability, validity, and efficiency. PHQ-2 is a quick, reliable, and valid screening tool to identify college students with depressive symptoms. Mass screening for depression among college students can improve surveillance and outreach. Early screening can enable timely treatment and counseling.	PHQ-2, PHQ-4, and GAD-2 were utilized as screening tools for depression and anxiety. Survey data were analyzed by using the SPSS for Windows statistical package 23.0.	Selection bias with a convenience sample. Recall bias, self-reported data, and inability to establish cause and effect relationships. Possible skewed responses to questionnaires by students.	Level VI Good Quality
4	(Wilson & Agius, 2017)	Integrative Review	N/A	PHQ-2 is a reliable and valid screening tool for detecting, not diagnosing, depression.	N/A	The search strategy was not identified in the paper.	Level V High Quality

5	(Soltani, Smith, Beck, & Johnson, 2014)	Retrospective Medical Record Review	N = 215 adult patients greater or equal to the age of 18 Two Student-Run Free Clinics ❑ N/A	Depression screening can be implemented within the electronic health record during routine primary care visits utilizing the PHQ-2 as an initial screening tool. If a patient scores equal to or greater than a 3, then it is protocol to administer the PHQ-9.	This study adopted a 2-stage screening strategy using the PHQ-2 and PHQ-9 . They queried the HER for patient demographics, visit dates, PHQ-2/PHQ-9 scores, and a diagnosis of depression on the problem list, PMH, or visit diagnoses fields. Data was analyzed using	Retrospective studies are more prone to bias. PHQ-2/PHQ-9 requires literacy for self-administration. Population was mainly Latino monolingual Spanish speakers. Thus, the population does not represent the general population.	Level III Good Quality

					Excel 14.2.5 and VassarStats.		
6	(Ajinkya, Schaus, & Deichen, 2016)	Well-Designed Non-Experimental Descriptive Study	N = 541 students University-Run Student Health Center	Housing arrangement was not associated with higher odds of depression among college students. However, a college student with a major in the College of Arts and Humanities was significantly associated with increased risk of depression. 13.1% of the study population tested positive for depression on the PHQ-9 and 56.3% were women.	College students were recruited and asked permission to take the PHQ-2 if they had an appointment at the Student Health Center for any reason. The students who scored positively on the PHQ-2 were then asked to take the PHQ-9 survey.	This study did not address the root cause of a positive association between depression and a major in the arts and humanities. Some the questions had missing data. Overall prevalence of depression may be biased because they aimed to have	Level III Good Quality

					Logistic regression was used to test the significance of associations between a few study variables and an outcome of depression. Chi-square tests or Fisher's exact test was used. Stata 11 was used to analyze data.	a certain number of cases for analysis. Study population was a convenience sample from the Student Health Center.	
7	(El-Den, Chen, Gan, Wong, & O'Reilly, 2018)	Systematic Review	N = 60 publications Studies were conducted in 24	There is no perfect screening tool for depression. PHQ-9 was the most evaluated tool. PHQ-2	PHQ-1, PHQ-1, PHQ-3, and other	This systematic review included studies regardless	Level I High Quality

			countries and 18 languages on 48,234 adults	was the second most evaluated tool in this systematic review . PHQ-2 has the ability to identify those with and without the disease.	screening tools were evaluated.	of methodological quality or design, which may limit generalizability and introduce bias.	
8	(Lipson, Lattie, & Eisenberg, 2019)	Descriptive Study	N = 155,026 college students from 196 college campuses ❑ N/A	In the past decade, there has been a substantial increase of mental health service utilization rates from 19% in 2007 to 34% in 2017. College students with lifetime diagnoses increased from 22% to 36%. The prevalence of depression and suicidality increased in the past decade as well. The study recommends universal	The study descriptively analyzed changes in mental health service utilization over the past decade among college students in the United States. The utilization rates reported the	This study could not determine the quality of mental services and if they conformed to evidence-based guidelines. Also, this study only utilized one item to measure perceived	Level III High Quality

				prevention programs and rapid referral to community treatment. Digital mental health program (apps or online) could expand treatment accessibility and minimize burden on service delivery systems.	proportion of all students in a given year with past-year therapy/counseling, past-year medication use, past-year treatment, and any lifetime diagnosis of a mental health condition. The study used Stata 14.2 to analyze data.	and personal stigma.	
9	(Kosyluk, Al-Khouja, Bink, Buchholz,	Single Randomized Control Trial	N = 198 college students	The findings of this study reveal that both contact- and education-based interventions	Participants were randomly assigned to a contact-based	Presentation/education may not have been relevant or	Level II High Quality

	Ellefson, Fokuo, … Corrigan, 2016)	Setting: Chicago University	showed a significant impact on personal stigma, perceptions of empowerment, discrimination, attitudes towards treatment seeking, and intentions to seek treatment from formal sources. This is useful information because it addresses the barriers to mental health care among college students.	anti-stigma presentation, education-based presentation, or control condition to evaluate if there is any difference towards stigma reduction.	tailored to college students. The study did not document the impact of the stigma change conditions after postintervention. The sample was not representative of the overall population of college students.	
10	(Slabaugh, Harris, & Wilcock, 2018)	Single Qualitative Study	N = 165 PHQ-2 were completed N = 8 participants (college students	This study demonstrated that the standardized depression screening was easily implemented without excessive	PHQ-2 was utilized as an initial screening tool to detect college	Participants were only screened for half of the semester and it may impact	Level V Good Quality

	aged 18-22) received interventions for positive screens Setting: College Health Center at a small, liberal arts Christian college	burden to staff or budget. The depression screening took approximately 2-7 minutes out of the visit depending if the depression screening was positive.	students with depression symptoms. If they scored positively, they were then administered the PHQ-9. If the patient scored between 10-19, then they encourage follow-up and gave the participant an educational handout. If the participant scored 20 or more OR was	the findings. This study only collected data from college students who had a nurse practitioner appointment, which can limit other students from being screened.

					possibly suicidal, then the nurse practitioner evaluated the participant immediately and treatment was initiated as appropriately.		
11	(Sandra, 2019)	Expert Opinion	N/A	This article offers recent statistics of depression and suicide rates among U.S. college students. In addition, the article offers recommendations to address the increased rates of depression in college students.	N/A	This study is based on an expert's opinion and it may be subjective, which may lead to bias.	Level V Good Quality

12	(Elkins, Plante, Germain, & Morley, 2017)	Comparative Cohort Study	N = 268 students Setting: Medical school	The findings suggest that students at risk for depression may be at higher risk of burnout too. As burnout increases, other traits such as empathy decreases, which can negatively impact patient outcomes.	This study measured the differences between MS1 and MS3 students at a medical school. PHQ-2 was utilized to screen for depression in medical students.	This study did not track burnout and depression longitudinally in individual students. Also, differences between MS1 and MS3 were not observed. The study was conducted at one public medical institution, which may not be generalized to other institutions.	Level II High Quality

13	(Maurer, Raymond, & Davis, 2018)	Expert Opinion	N/A	The article discusses different screening tool that could be utilized to detect depression and compares the sensitivity and specificity of each screening tool. The article was up to date statistics regarding depression in the United States.	N/A	This study is based on an expert's opinion and it may be subjective, which may lead to bias.	Level V Good Quality
14	(Lakkis & Mahmassani, 2015)	Literature Review	N/A	This lit review contains an updated comprehensive document that overviews the existing depression screening recommendations and instruments specifically available for primary care providers.	N/A	Some references of the literature review are not current (within the last 5 years).	Level V High Quality

15	(Lipson, Gaddis, Heinze, Beck, & Eisenberg, 2015)	Cross-Sectional Study	N = 43,210 undergraduates at 72 college campuses in the U.S.	The results revealed that the following institutional characteristics are associated with worse mental health: doctoral-granting, public, large enrollment, nonresidential, less competitive, and lower graduation rates. There are substantial variations in student mental health and treatment utilization across U.S. colleges.	This study focused on associations between institutional characteristics, student mental health, and treatment utilization.	The overall survey response rate was under 30%. The sample was composed of colleges that chose to participate in the Healthy Minds Study.	Level IV High Quality
16	(Wright et al., 2015)	Descriptive Study	N = 4010 adolescents enrolled in a large integrated care system	Adolescents experiencing depressive symptoms are associated with increased health care utilization and costs. Depression screening and	PHQ-2 and PHQ-9 were utilized to screen adolescents for depression. Chi-square, Wilcoxon	Data were from a single health care system only. This study cannot make determinations	Level III High Quality

				evidence-based mental health services can assist with better health care costs control.	rank sum, and *t* tests were used to test for statistical differences in outcomes.	about causalities of relationships between medical problems and the incidence of depressive symptoms.	
17	(Aalsma et al., 2017)	Prospective Cohort Study	N = 2,038 adolescents within two clinics in Indiana	A majority of healthcare providers used screening results to guide clinical care when a computer-based decision support system algorithm focused on depression screening was implemented. Thus, depression screening should be integrated in EMR/EHR.	PHQ-2 and PHQ-9 were used to screen adolescents for depression in primary care.	The study had no way of determining if the participants completed the questionnaires on their behalf or if their parents helped. The generalizability of	Level II High Quality

						the study's findings may be limited since it was implemented into one health-care system.	
18	(Dannenberg et al., 2019)	Multiphasic Mixed Methods Study	N = 6 focus groups with 15 participants, 7 clinician interviews, and 10 usability sessions	This study demonstrated acceptability of the process of depression screening linked to decision support via an electronic tablet in primary care waiting rooms.	Observable measures include focus groups of patients with depression, interviews with primary care clinicians, and usability sessions where patients used	Several focus groups had fewer participants than expected. Study participants were mainly Caucasian, limiting the relevance of the findings to a more diverse population.	Level III Good Quality

					the current version of encounter decision aid in a primary care waiting room.		
19	(Etter et al., 2017)	Prospective Cohort Study	N = 2,134 adolescent patients aged 12-20 in 2 primary care clinics in Indianapolis, Indiana	This study demonstrated that computer decision support systems can be utilized effectively in primary care to screen adolescents for suicidality and provide follow-up recommendations for healthcare providers.	The Child Health Improvement through Computer Automation (CHICA) system was used to integrate EMR system data, pre-visit screening data, and correlative	This study does not assess recency of suicidality. This study did not assess whether the patients benefitted from suicide screening procedures or provider follow-up.	Level II High Quality

					provider responses from previous visits.		
20	(Ennis et al., 2019)	Longitudinal Cohort Study	N = 392 new college entrants who all reported some lifetime mental disorder or suicidal thoughts or behaviors Setting: Northern Ireland	The findings of this study have implications for the active screening and intervention for vulnerable college students, especially males and those with mental disorders.	Suicidal though, plans, and behaviors/attempts were assed using the Self-Injurious Thoughts and Behavior Interview (SITBI). Alcohol disorders were screened by the Alcohol Use Disorders Identification Test (AUDIT). ADHD	The findings are correlational and causal relationships cannot be inferred. Sample size did not allow consideration of several sub-groups.	Level IV Good Quality

					was assessed via the ADHD Self-Report Scale.		
21	(English & Campbell, 2019)	Cross-Sectional Study	N = 131 college students at public 4-year universities	The findings revealed that universal depression screening is common among college health centers.	The study deployed surveys via Qualtrics, which is an online survey. The study assessed college health center demographics and health screening practices, including if the clinics reports to universally screen for	There were limitations to the representativeness of the current sample. Selection bias may have played a part in the present results.	Level IV Good Quality

					depression in college students.		
22	(Smith, Brown Levey, & Lyon, 2017)	Nonrandomized Quasi-Experimental Study	N = 5 medical assistants and 2 healthcare providers Setting: Family Medicine Residency as intervention group and UCHealth Internal-Medicine Residency as control group	The Ambulatory Process Excellence (APEX) Model is a team-based primary care delivery model and contributes to better depression screening processes. Other primary care offices could implement the APEX Model and improve depression screening processes, despite increasing new-patient and overall-visit volumes.	Covariates such as monthly visits, new patient appointments, PHQ-2 screening rates, patients screening positive, and PHQ-9 screening rates were measured. All data was transformed into modified *z* scores on Microsoft Excel, and the scores were plotted	The intervention setting practiced with a culture of innovation, which may risk volunteer bias. Control charts using standard scores can lead to interpretive errors analogous to conflating statistical and clinical significance.	Level II High Quality

					on a multivariate control chart for each practice. Then, findings were analyzed for evidence of special cause variation.		
23	(Manea et al., 2016)	Systematic Review with Diagnostic Meta-Analysis	N/A	PHQ-2 cutoff at ≥ 2 may be preferred to ensure that few cases of depression are missed. However, in situations in which the depression prevalence is low, the cutoff at ≥ 2 may result in an unacceptably high false-positive rate because of the associated modest specificity.	N/A	There is a possibility of publication bias. Only one author was assigned to study selection and data extraction, which can introduce bias.	Level I High Quality

24	(Becker, Dvorsky, Holdaway, & Luebbe, 2018)	Cross-Sectional Study	N = 1700 college students aged 18-29 from two universities in the Midwestern U.S.	Sleep problems was significantly associated with suicidal behaviors above and beyond depressive symptoms. Suicide risk was 6.54 times greater for participants with elevated depressive symptoms and 2.7 times greater for participants with sleep problems.	Sleep was measured by the Pittsburgh Sleep Quality Index (PSQI) and depression was measured with the Depression Anxiety Stress Scales-21 (DASS-21). Suicidal behaviors were measured with the Suicidal Behaviors Questionnaire-Revised (SBQ-R).	The study only included self-report rating scale measures and it can contribute to mono-informant biases. The sample from this study was disproportionately female and White.	Level IV High Quality

25	(Hill, Yaroslavsky, & Pettit, 2015)	Longitudinal Cohort Study	N = 262 college students	The findings can inform approaches to identify college students who are high-risk for depressive symptoms based on the presence of social disconnection and/or negative feedback-seeking.	Depressive symptoms were measured with the CES-D and reassurance seeking behaviors were measured with the Reassurance Seeking Scale. Negative feedback seeking was screened with the Negative Feedback Seeking Scale. Social disconnection was	The population was mainly female and Hispanic. The four-month spacing of assessments could have resulted in a failure to identify those who experience brief recurrent depressive episodes.	Level IV Good Quality

			assessed with the Interpersonal Needs Questionnaire. Rumination was assessed with the Ruminative Responses Scale. The study identified courses of depressive symptoms over a 12-month period among college students who screened positive on

					a depression symptom measure.		
26	(Ebert et al., 2019)	Prospective Cohort Study	N = 2,519 freshmen college students at baseline and 958 college students at one-year follow-up Setting: One Belgian College	The study's findings highly recommend depression screening at college entrance to identify students at risk for major depressive disorder (MDD). As a result, it may improve the development and deployment of targeted preventive interventions.	Socio-demographic variables, parental psychopathology, traumatic experiences in childhood to adolescence, stressful events experienced in the past year and a 12-month mental disorder were assessed, measured, and analyzed.	Response rates were only moderate. Baseline risk for mental disorders was not assessed by diagnostic interviews. The findings might not generalize to college students from other universities because the study	Level II High Quality

							was conducted among freshmen in one Belgian college.	
27	(Chan, Moore, Derenne, & Fuchs, 2019)	Expert Opinion	N/A	The article describes the challenges of the youth in navigating into life transition and offers recommendations to promote a successful transition into adulthood. Young adults who have a history of mood disorders such as MDD are uniquely vulnerable and may experience relapsing depression and are at risk for suicide.	N/A		This study is based on an expert's opinion and it may be subjective, which may lead to bias.	Level V Good Quality

Table A3: DNP Project Synthesis Matrix

Concept, Theme or Category	Source #1 Author	Source #2 Author	Source #3 Author	Source #4 Author
Main Idea A Universal Depression Screening vs Screening Those at High-Risk	Prediction of major depressive disorder onset in college students (Ebert et al., 2018)	Prevalence and characteristics of universal depression screening in U.S. college health centers (English & Campbell, 2019)	Enhancing depression screening to identify college students at risk for persistent depressive symptoms (Hill, Yaroslavsky, & Pettit, 2015)	Burnout and Depression in MS1 and MS3 Years: A Comparison of Cohorts at One Medical School (Elkins, Plante, Germain, & Morley, 2017)
Main Idea B Utilization of EMR for Routine Depression Screening	Suicide Screening in Primary Care: Use of an Electronic Screener to Assess Suicidality and Improve Provider Follow-Up for Adolescents (Etter et al., 2017)	End-user views of an electronic encounter decision aid linked to routine depression screening (Dannenberg et al., 2019)	Physician Intervention to Positive Depression Screens Among Adolescents in Primary Care (Aalsma et al., 2018)	

Main Idea C Initial PHQ-2 Screening Tool and then further screening with PHQ-9 if positive results	Evaluating Transformation with Available Resources: The Influence of APEX on Depression Screening (Smith, Brown Levey, & Lyon, 2017)	Screening instruments for depression in primary care: A concise review for clinicians (Lakkis & Mahmassani, 2015)	Association of adolescent depressive symptoms with health care utilization and payer-incurred expenditures (Wright, Katon, Ludman, McCauley, Oliver, Lindenbaum, & Richardson, 2016)	Universal depression screening, diagnosis, management, and outcomes at a student-run free clinic (Soltani, Smith, Beck, & Johnson, 2015)
Main Idea D PHQ-2, PHQ-4, & PHQ-9	Identifying depression with the PHQ-2: A diagnostic meta-analysis (Manea et al., 2016)	The psychometric properties of PHQ-4 depression and anxiety screening scale among college students	The psychometric properties of depression screening tools in primary healthcare settings: A systematic review (El-Den, Chen, Gan, Wong, & O'Reilly, 2018)	Initiation of Standardized Depression Screening in College Health: A Quality Improvement Project (Slabaugh, Harris, & Wilcock, 2018).

		(Khubchandani, Brey, Kotecki, Kleinfelder, & Anderson, 2016).		
Main Idea E Depression Screening and Depression Diagnosis	Depression: Screening and Diagnosis (Maurer, Raymond, & Davis, 2018)	Addressing the increase in depression in college students (Sandra, 2019)	Universal depression screening, diagnosis, management, and outcomes at a student-run free clinic (Soltani, Smith, Beck, & Johnson, 2015)	Initiation of Standardized Depression Screening in College Health: A Quality Improvement Project (Slabaugh, Harris, & Wilcock, 2018).
Main Idea F Risk Factors of Depression Among College Students	Variations in Student Mental Health and Treatment Utilization Across US Colleges and Universities (Lipson, Gaddis, Heinze, Beck, & Eisenberg, 2015)	Transitional age youth and college mental health (Chan, Moore, Derenne, & Fuchs, 2019)	Prediction of major depressive disorder onset in college students (Ebert et al., 2018)	Depression in health college students: relationship factors and sleep quality (Çelik, Ceylan, Ünsal, & Çağan, 2019)

Main Idea G Primary Healthcare Provider (PCP) Barriers to Depression Screening and PCP Education Regarding Depression Screening	Universal depression screening, diagnosis, management, and outcomes at a student-run free clinic (Soltani, Smith, Beck, & Johnson, 2015)		
Main Idea H Depression Symptoms (e.g. suicidal ideation)	Sleep problems and suicidal behaviors in college students (Becker, Dvorsky, Holdaway, & Luebbe, 2018)	Predictors of depressive symptoms in college students: A systematic	Readiness to change and barriers to treatment seeking in college students with a mental disorder

Among College Students		review and meta-analysis of cohort studies (Liu et al., 2019)	(Ennis et al., 2019)
Main Idea I Mental Illness Stigma Among College Students	Challenging the Stigma of Mental Illness Among College Students (Kosyluk et al., 2016)	Addressing the increase in depression in college students (Sandra, 2019)	Increased rates of mental health service utilization by U.S. college students: 10-year population-level trends (2007–2017) (Lipson, Lattie, & Eisenberg, 2019)

Appendix B

Figure B1: Rogers' Diffusion of Innovations Theory

Rogers' Diffusion of Innovations Theory		
Phase 1	Awareness	Awareness and knowledge of the innovation.
Phase 2	Interest	If there is interest in the innovation by the staff, the change process progresses.
Phase 3	Evaluation	The innovation will be evaluated by the staff and will be either accepted or rejected.
Phase 4	Trial	The innovation can be implemented in part or in whole, and may be evaluated at intervals.
Phase 5	Adoption	The final phase in which the innovation will be adopted or rejected. If the innovation is rejected, then it may be considered at a later time.

(Orr & Davenport, 2015)

Appendix C

Figure C1: Patient Health Questionnaire-2

Use: The purpose of the PHQ-2 is not to establish a final diagnosis or to monitor depression severity, but rather to screen for depression as a "first-step" approach.

Scoring: A PHQ-2 score ranges from 0 to 6; patients with scores of 3 or more should be further evaluated with the PHQ-9, other diagnostic instrument(s), or a direct interview to determine whether they meet criteria for a depressive disorder.

Patient Name: _____ Date of Visit: _____

Over the past 2 weeks, how often have you been bothered by any of the following problems?	Not at all	Several days	More than one-half of the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3

Figure C2: Patient Health Questionnaire-9

Patient Name: _____ Date of Visit: _____

Over the past 2 weeks, how often have you been bothered by any of the following problems?	Not at all	Several days	More than one-half of the days	Nearly every day
1. Little interest or pleasure in doing things	0	1	2	3
2. Feeling down, depressed, or hopeless	0	1	2	3
3. Trouble falling asleep, staying asleep, or sleeping too much	0	1	2	3
4. Feeling tired or having little energy	0	1	2	3
5. Poor appetite or overeating	0	1	2	3
6. Feeling bad about yourself—or that you're a failure or have let yourself or your family down	0	1	2	3
7. Trouble concentrating on things, such as reading the newspaper or watching television	0	1	2	3
8. Moving or speaking so slowly that other people could have noticed; or the opposite—being so fidgety or restless that you have been moving around a lot more than usual	0	1	2	3
9. Thoughts that you would be better off dead or of hurting yourself in some way	0	1	2	3

10. If you checked off any problems listed above, how difficult have those problems made it for you to do your work, take care of things at home, or get along with other people? ❒ Not difficult at all ❒ Somewhat difficult ❒ Very difficult ❒ Extremely difficult

Appendix D

Figure D1: Depression Screening Protocol Process Map for Pilot Period

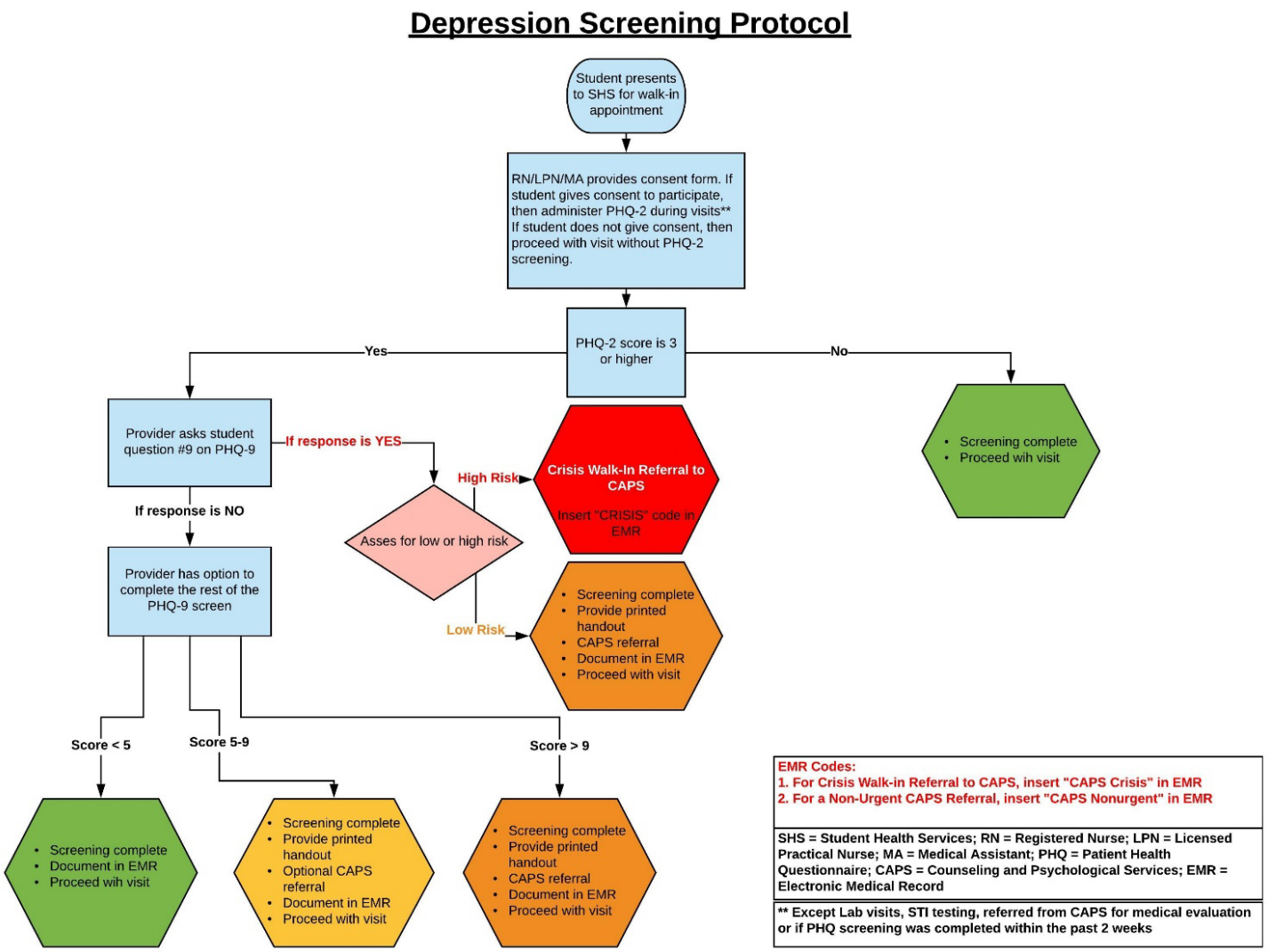
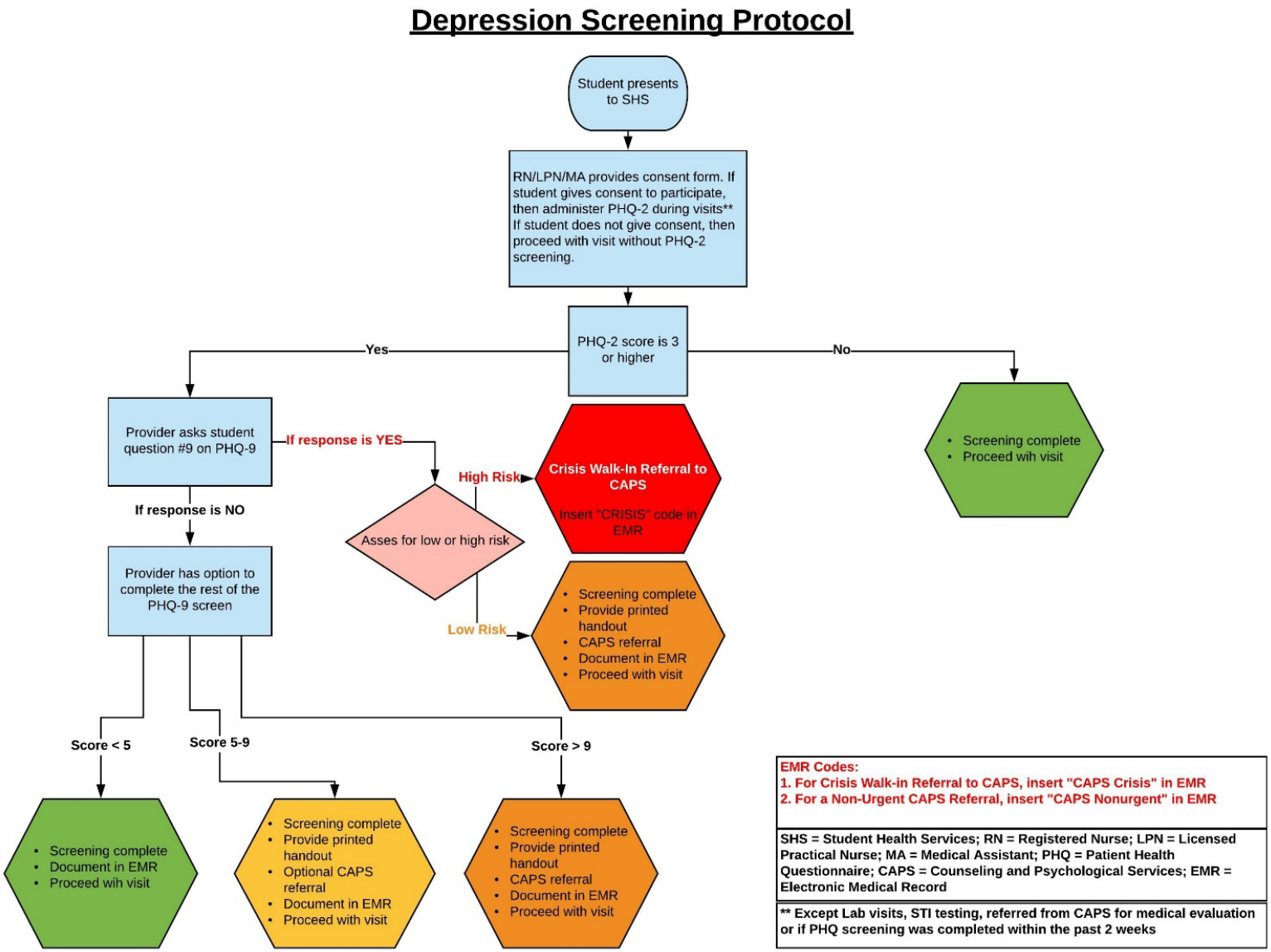


Figure D2: Depression Screening Protocol Process Map for Two-Month Implementation Period



Appendix E

Figure E1: Educational Handout - Front Side

STUDENT HEALTH SERVICES

WHAT IS DEPRESSION?

Depression is a common illness that impacts how you feel, think and act.
Depression can negatively impact your functioning at home and at school.

Depression symptoms vary from mild to severe and may include:

- Feeling sad and/or numb
- Increase in irritabilty
- Loss of interest or pleasure in activities you used to enjoy
- Changes in appetite or weight
- Trouble sleeping or sleeping too much
- Loss of energy and fatigue
- Increase in agitation (e.g., pacing) or slowed movements/speech
- Feeling worthless or guilty
- Difficulty thinking, concentrating or making decisions
- Thoughts of death or suicide

HOW CAN I TREAT IT?

MOVE

Engage in physical activity, especially if all you want to do is stay in bed. Spending time in nature can also help.

TALK

People who are depressed often isolate from friends and family. Reaching out helps break the cycle!

BE AWARE

Depression can be like wearing "negativity glasses" because it makes everything seem hopeless. Understanding this can help you identity your own negative bias.

ENGAGE

Do something you enjoy, even if you don't feel like it. Activies that are meaningful to you can be especially helpful.

AVOID DRUGS AND ALCOHOL

Many people who feel depressed use alcohol, cannabis, and other substances to feel better. However, substance can make your symptoms worse and cause you to act impulsively.

Figure E2: Educational Handout - Back Side

WHERE CAN I GO FOR HELP?

ON YOUR PHONE, ANYTIME, FREE

TAO
Therapy Assistance Online

Therapy Assisted Online (TAO), is an interactive resource you can sign up for at thepath.taoconnect.org with your FGCU email. Once you register, you can download the app. on your phone, Learn more about depression and how you can cope!

ON CAMPUS, CONFIDENTIAL, FREE

CAPS
Counseling and Psychological Services

CAPS will assist you in coming up with a plan and connecting you to the appropriate resources. You can schedule an appointment by coming to the 3rd floor of Student and Community Counseling. Call 239-590-7950 for more information. You can also visit fgcu.edu/caps for more information about depression, including self help resources.

CAPS also has a 24/7/365 Help Line: (239)-745-EARS

HEALTH INSURANCE

Many insurance companies cover mental health treatment. Call and ask your insurance company for details. You can also visit the CAPS website and view the **Community Provider Database** to find a provider nearby.

Having thoughts of suicide?

NATIONAL SUICIDE PREVENTION LIFELINE™
1-800-273-TALK
www.suicidepreventionlifeline.org

Reach out to someone - a friend, family member, or a mental health professional - and tell them how you've been feeling. You are not alone.

The Suicide Prevention Lifeline or the Crisis Text Line are both safe, non-judgmental options to talk about your thoughts.

In Crisis?
Text HELLO to 741741
CRISIS TEXT LINE | Free, 24/7, Confidential

If you are ever in immediate danger of hurting or killing yourself, you can call 911 or go to the nearest hospital to stay safe.

Appendix F

Figure F1: Pilot Period Age Results

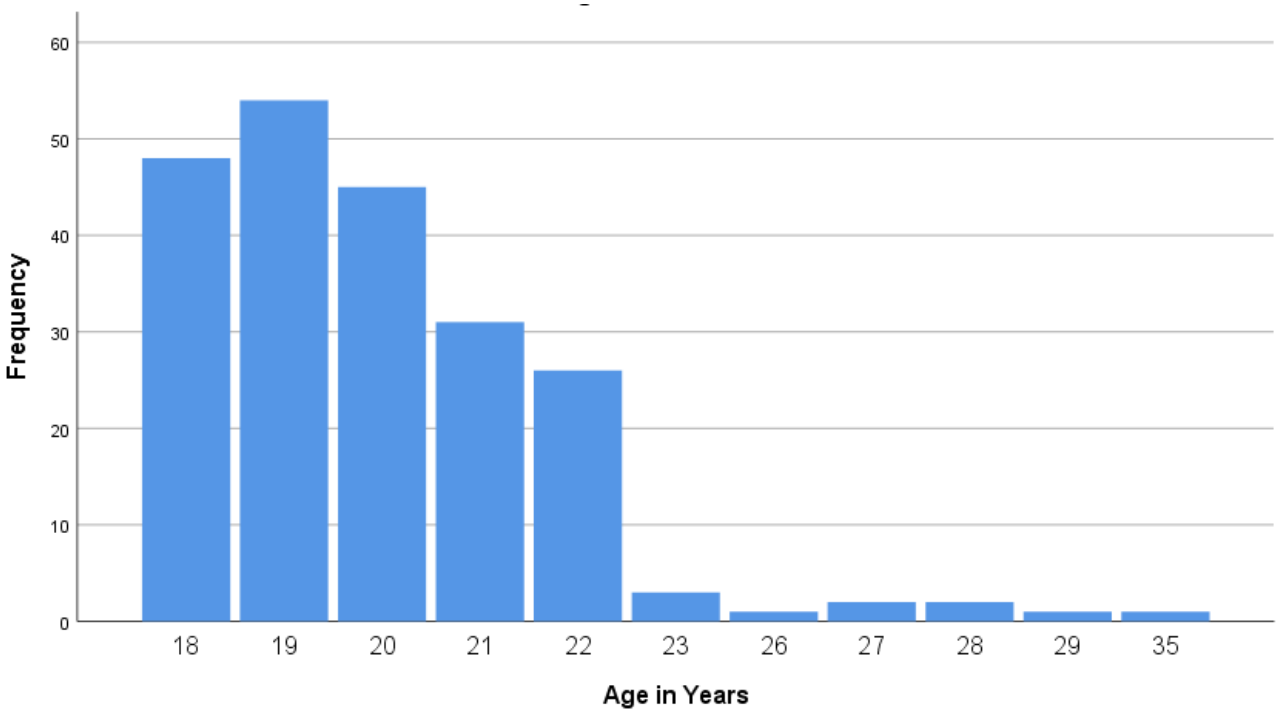


Figure F2: Pilot Period Gender Results

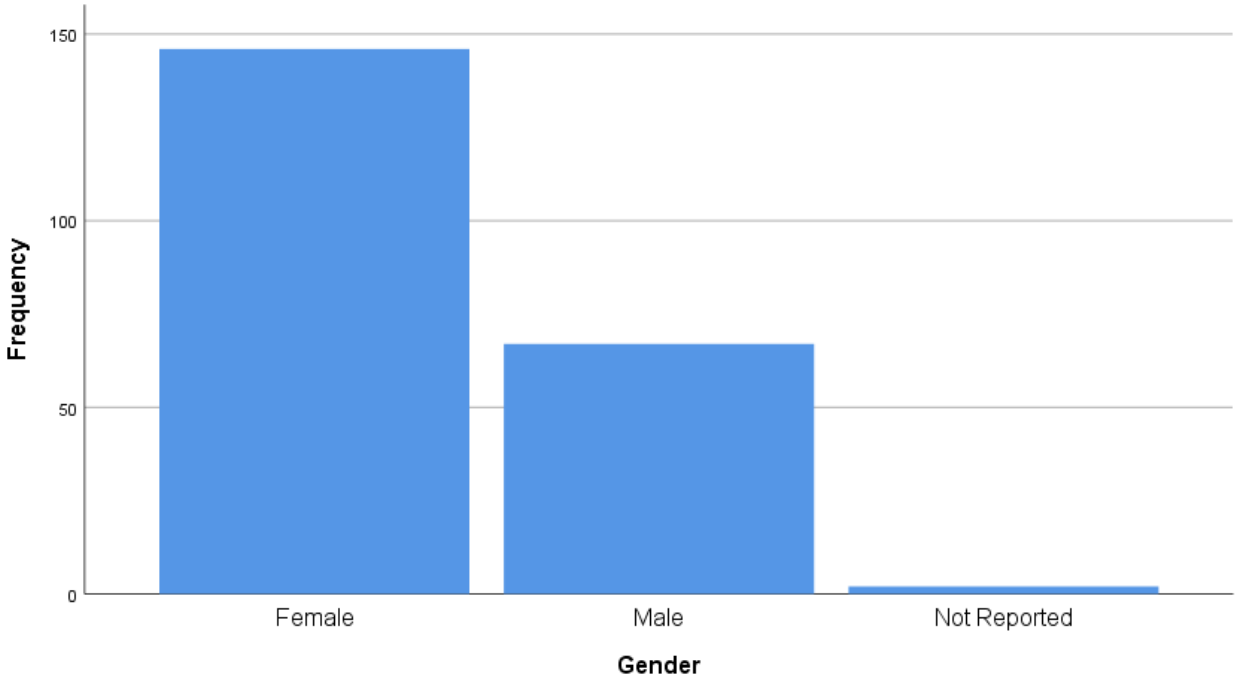


Figure F3: Pilot Period Ethnicity Results

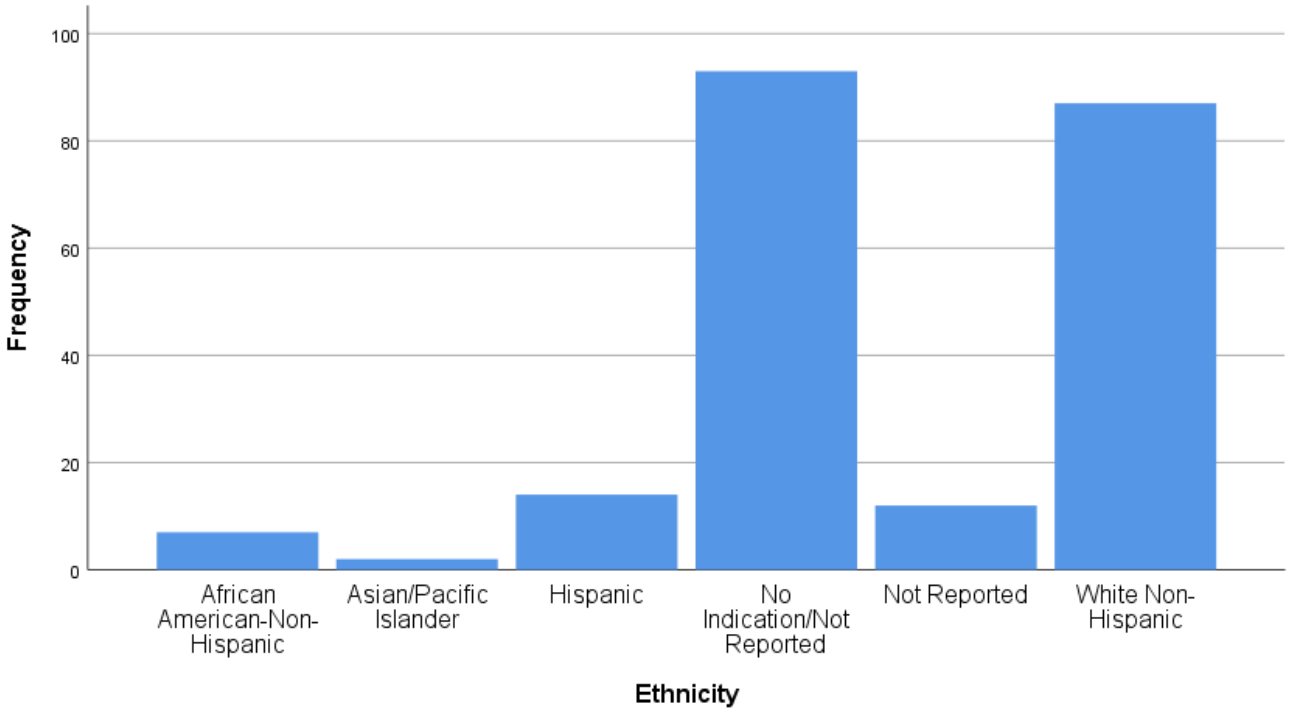


Figure F4: Pilot Period PHQ-2 & PHQ-9 Frequency Results

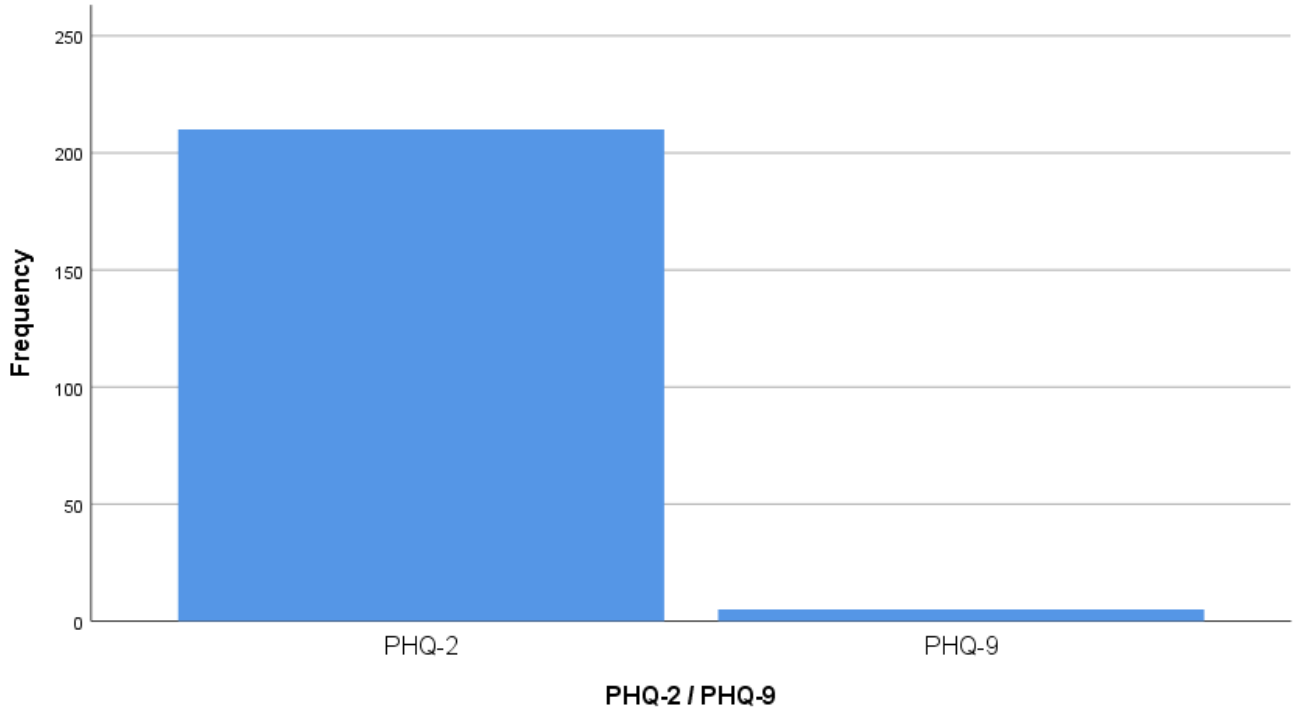
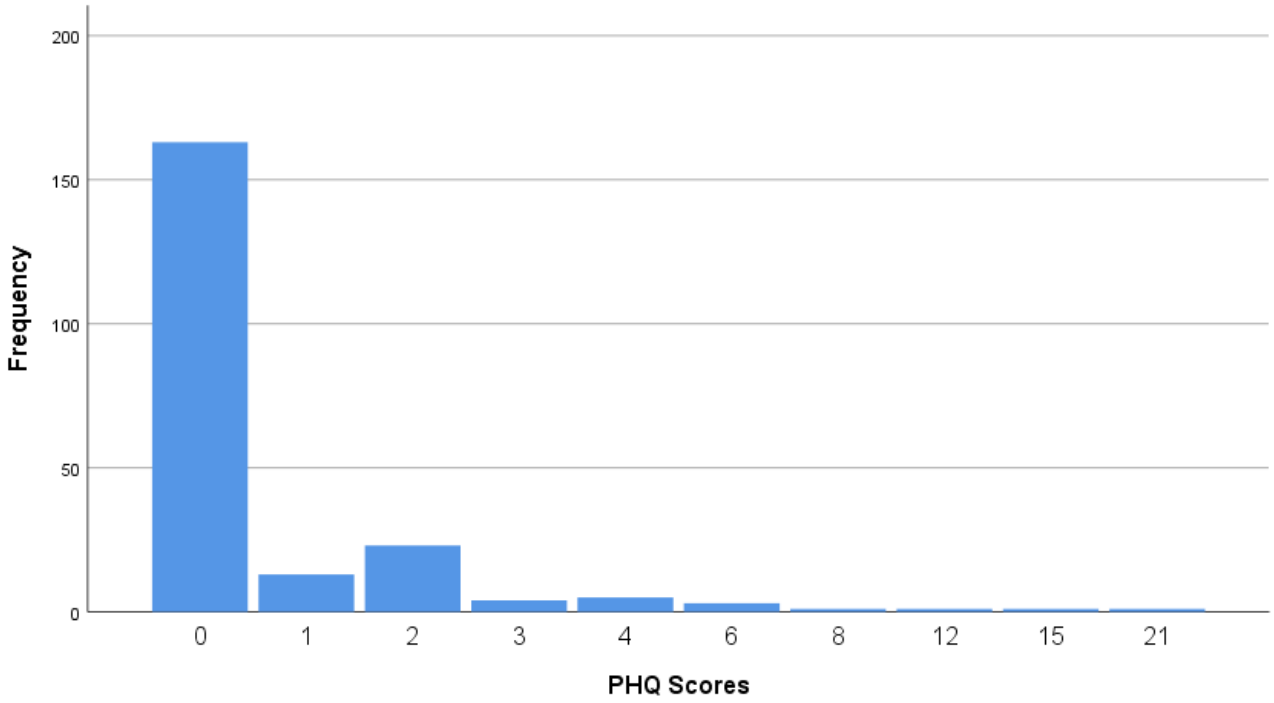


Figure F5: Pilot Period PHQ Total Scores



Appendix G

Figure G1: Two-Month Implementation Period Age Results

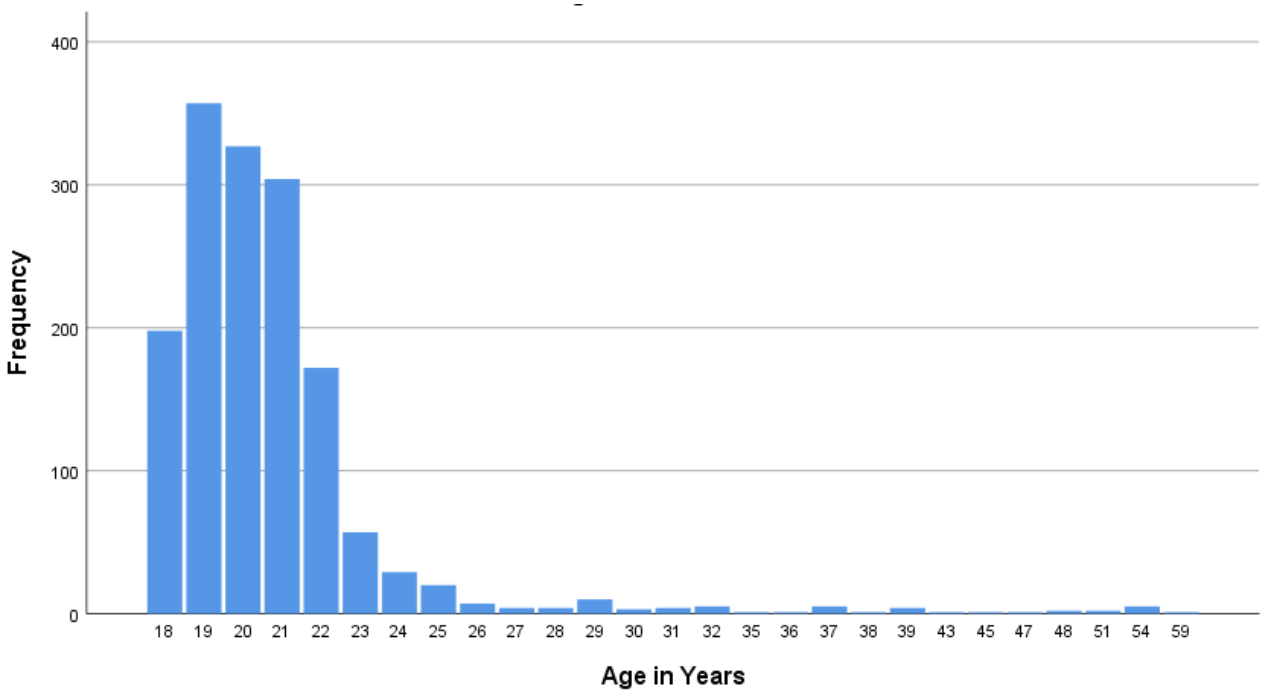


Figure G2: Two-Month Implementation Period Gender Results

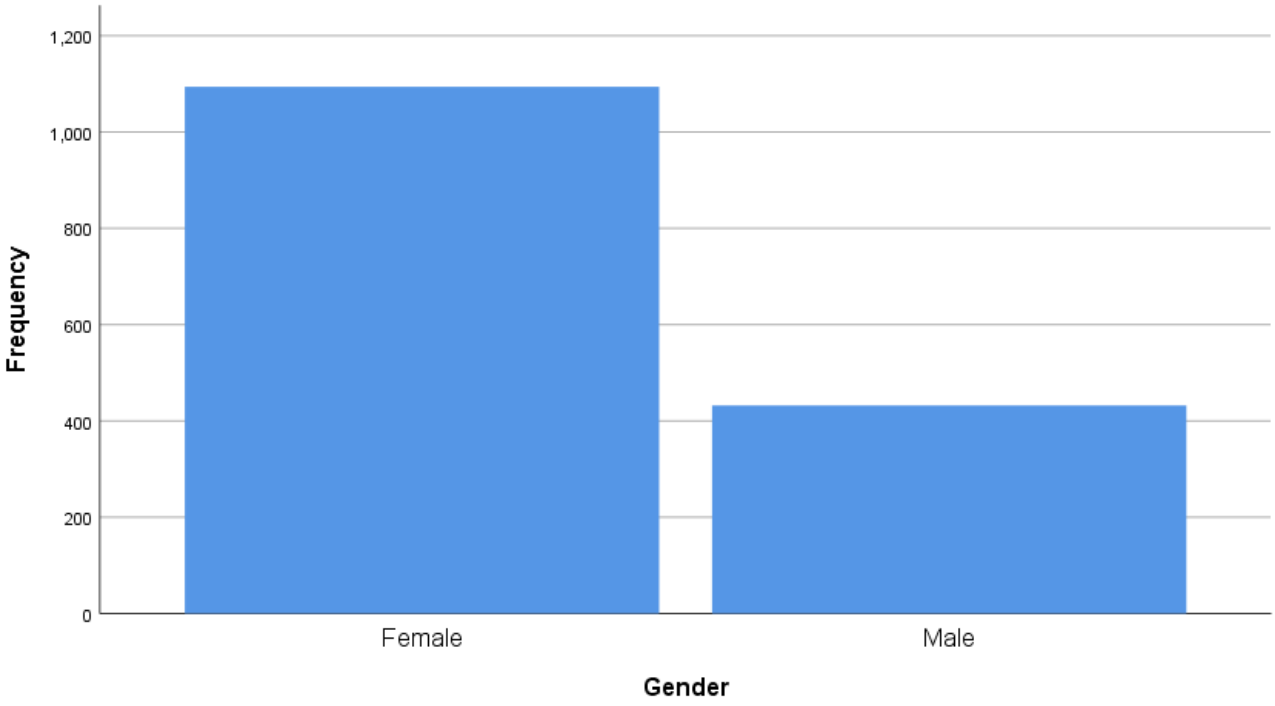


Figure G3: Two-Month Implementation Period Ethnicity Results

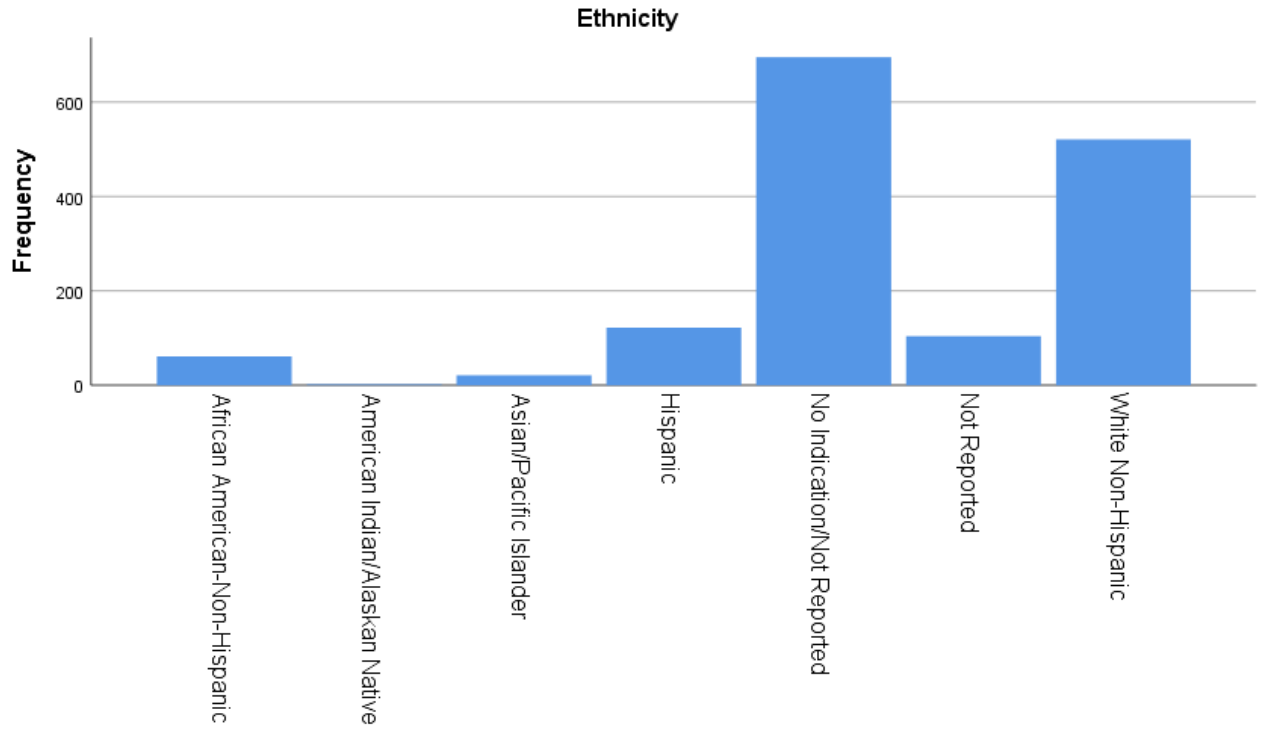


Figure G4: Two-Month Implementation Period PHQ-2 & PHQ-9 Frequency Results

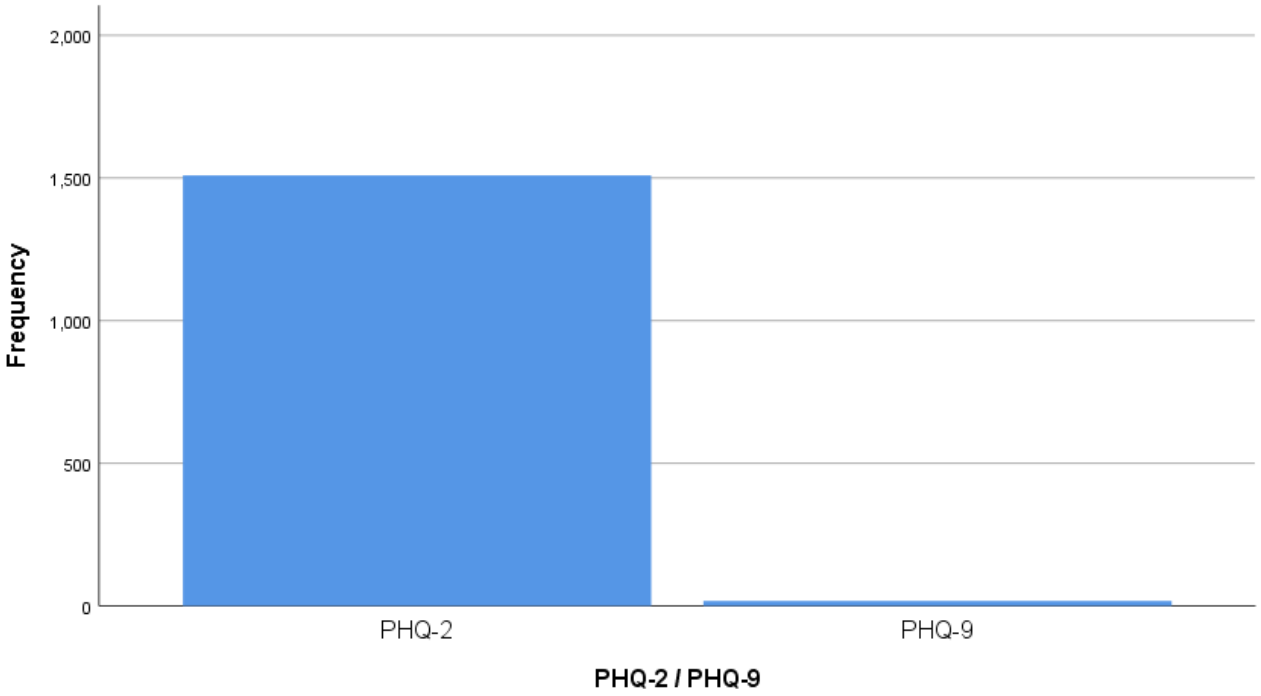


Figure G5: Two-Month Implementation Period PHQ Total Scores

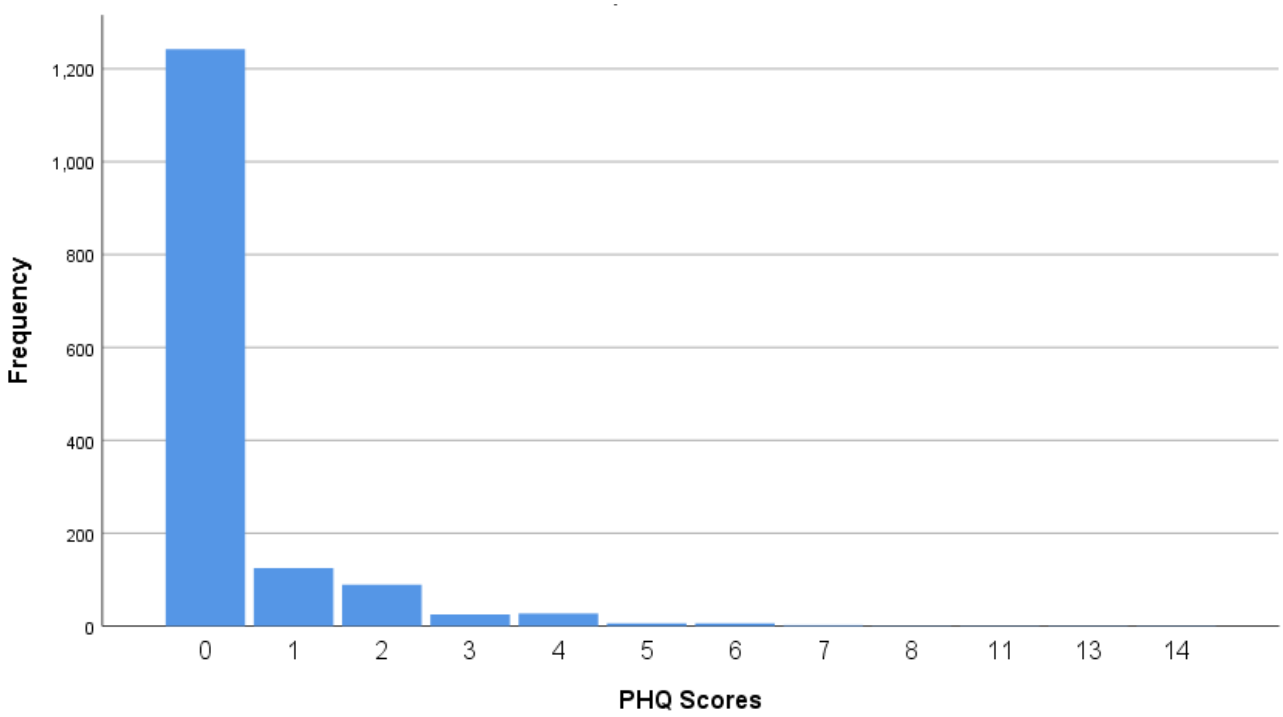


Figure G6: Two-Month Implementation Period Encounter Date Frequencies

